Optimization and Regularization of Neural Networks

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Outline

Optimization

Regularization of NN

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Optimization

- Challenges in Optimization
- Momentum
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- Dropout

Learning vs. Optimization

Goal of learning: minimize generalization error, or the loss function

$$\mathcal{L}(\boldsymbol{W}) = \mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim \boldsymbol{p}_{data}}\left[L(f\left(x, W\right), y\right]$$

f is the neural network

In practice, empirical risk minimization:

$$\mathcal{L}(W) = \sum_i [L(f(x_i; W), y_i]$$

Quantity optimized different from the quantity we care about

Local Minima

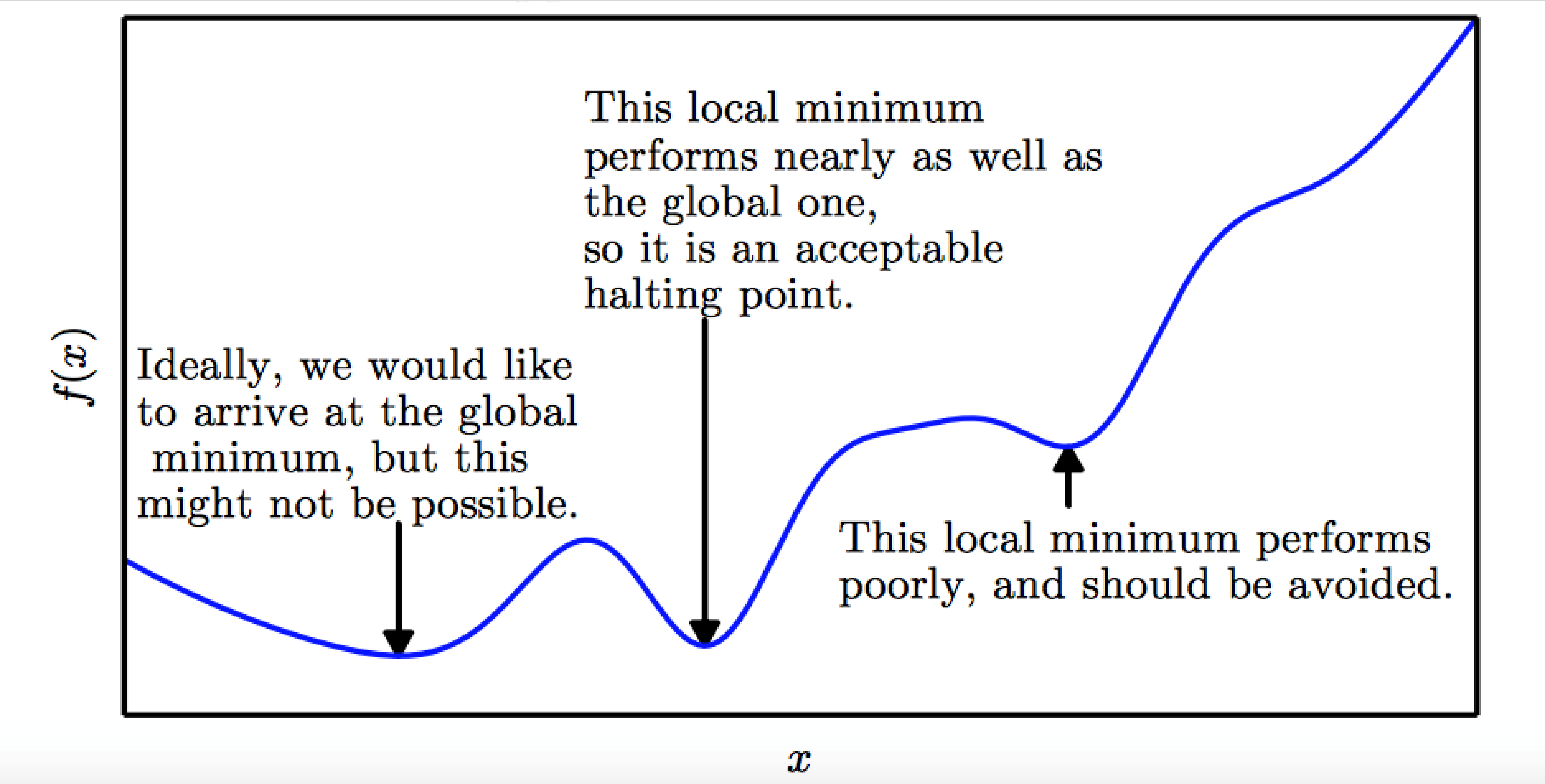


Image Source: Goodfellow et al. (2016)

Critical Points

Points with zero gradient

2^{nd}-derivate (Hessian) determines curvature

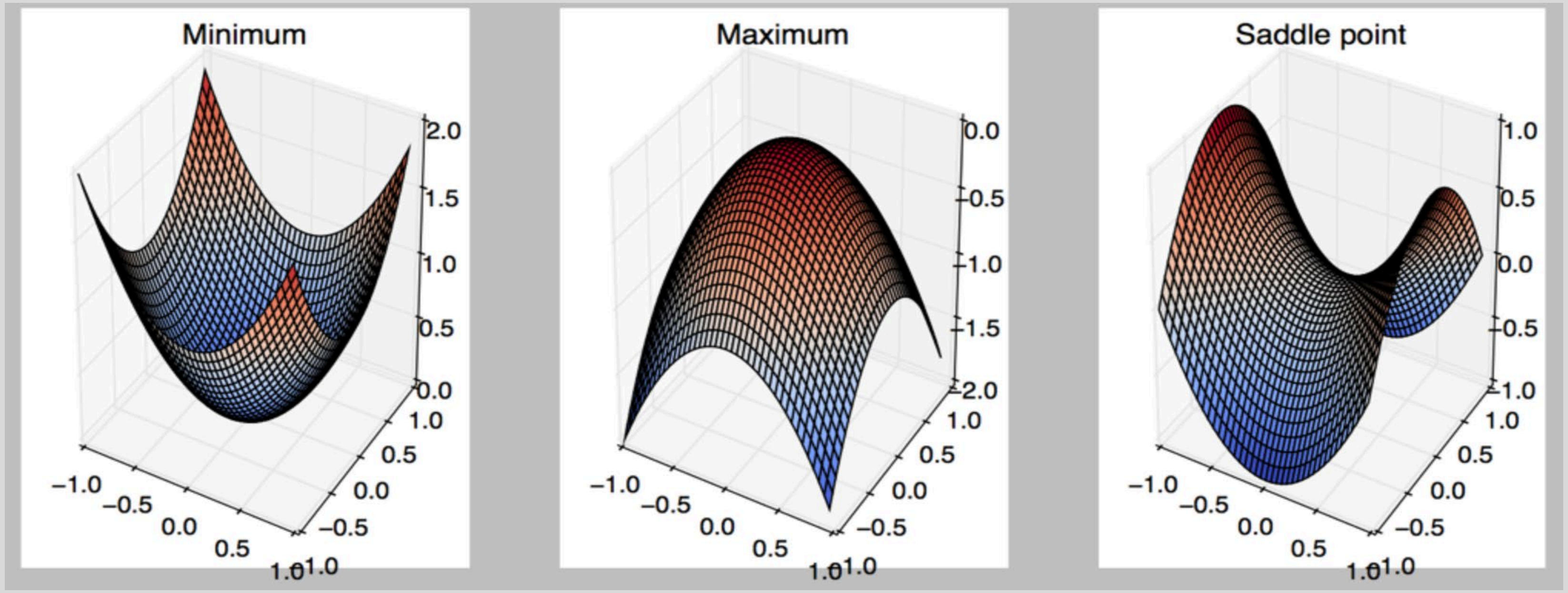


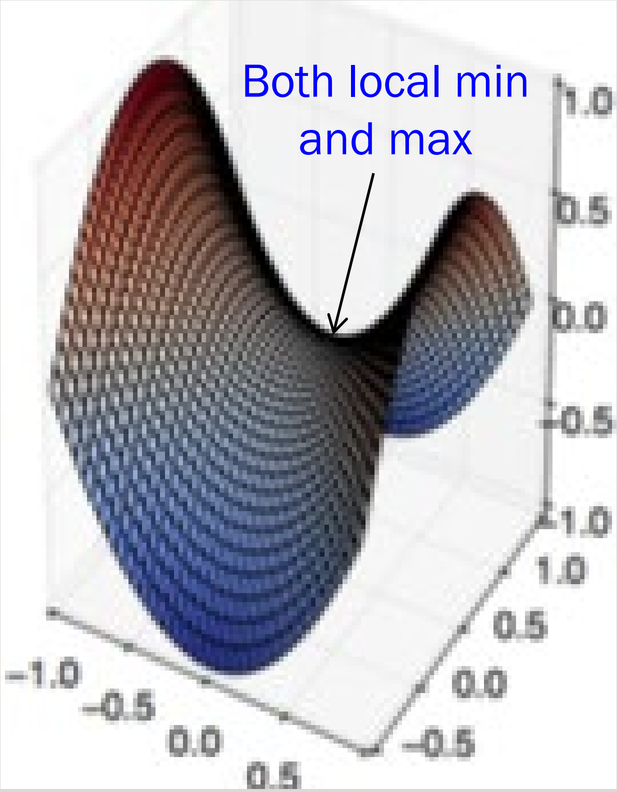
Image Source: Goodfellow et al. (2016)

Local Minima

Old view: local minima is major problem in neural network training

Recent view:

- For sufficiently large neural networks, most local minima incur low cost
- Not important to find true global minimum



Saddle Points

Recent studies indicate that in high dim, saddle points are more likely than local min

Gradient can be very small near saddle points

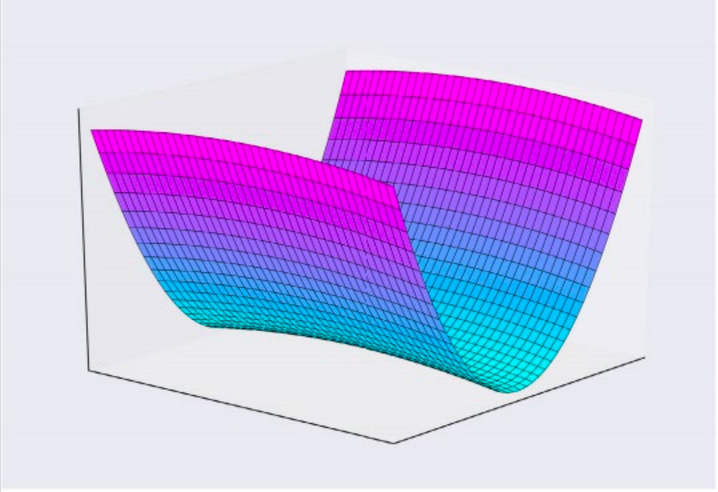


Image Source: Goodfellow et al. (2016)

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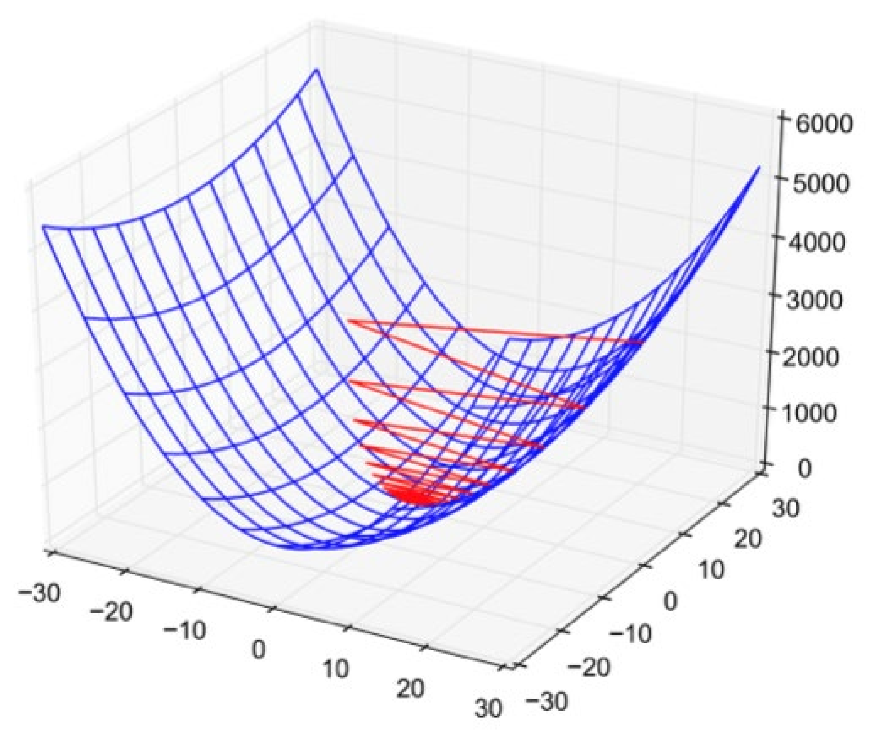
Poor Conditioning

Poorly conditioned Hessian matrix

- High curvature: small steps leads to huge increase

Learning is slow despite strong gradients

Oscillations slow down progress



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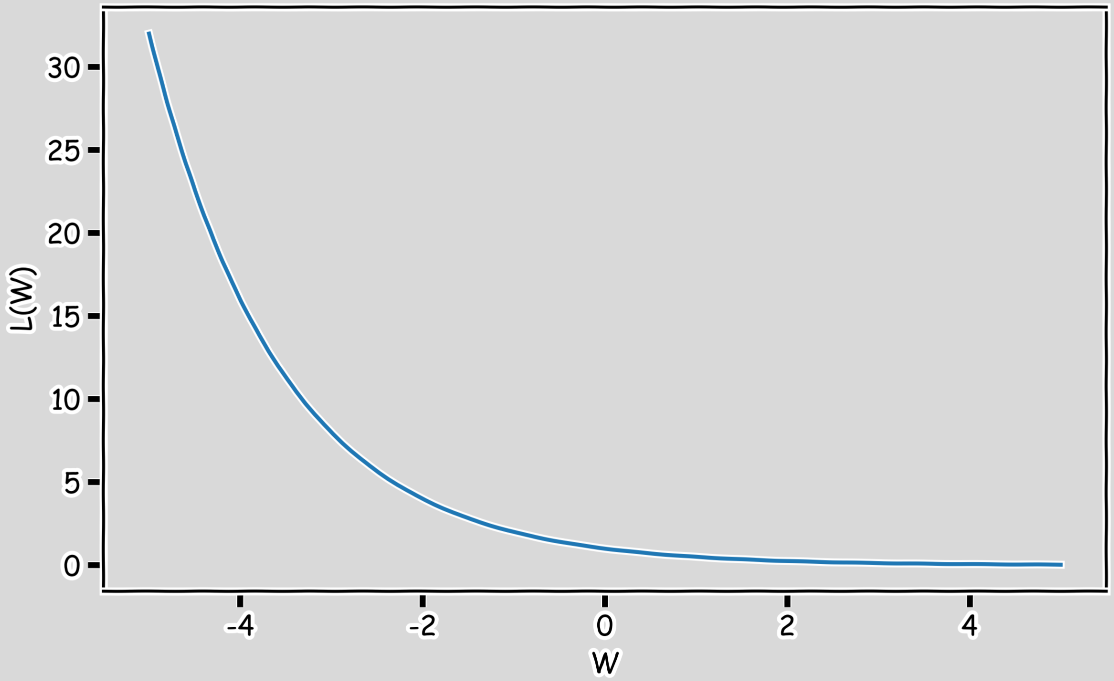
No Critical Points

Some cost functions do not have critical points. In particular: classification.

WHY?



Exploding and Vanishing Gradients

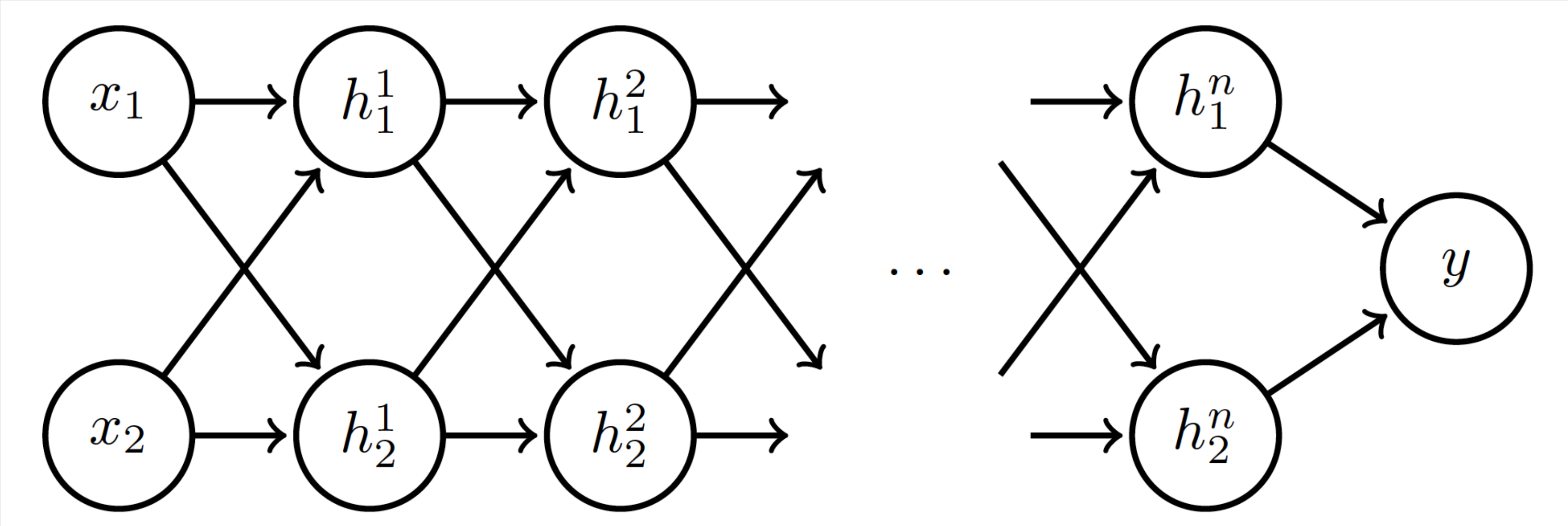


Image Source: deeplearning.ai

Linear activation $h_i = Wx$

$$h_i = Wh_{i-1}, \qquad i = 2, \ldots, n$$

Exploding and Vanishing Gradients

$$\text{Suppose } \mathbf{W} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}:$$

$$\begin{bmatrix} h_1^1 \\ h_2^1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \cdots \quad \begin{bmatrix} h_1^n \\ h_2^n \end{bmatrix} = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Exploding and Vanishing Gradients

$$\text{Suppose } x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Case 1: } a = 1,\ b = 2:$$

$$y \to 1, \quad \nabla y \to \begin{bmatrix} n \\ n2^{n-1} \end{bmatrix}$$

Explodes!

$$\text{Case 2: } a = 0.5,\ b = 0.9:$$

$$y \to 0, \quad \nabla y \to \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Vanishes!

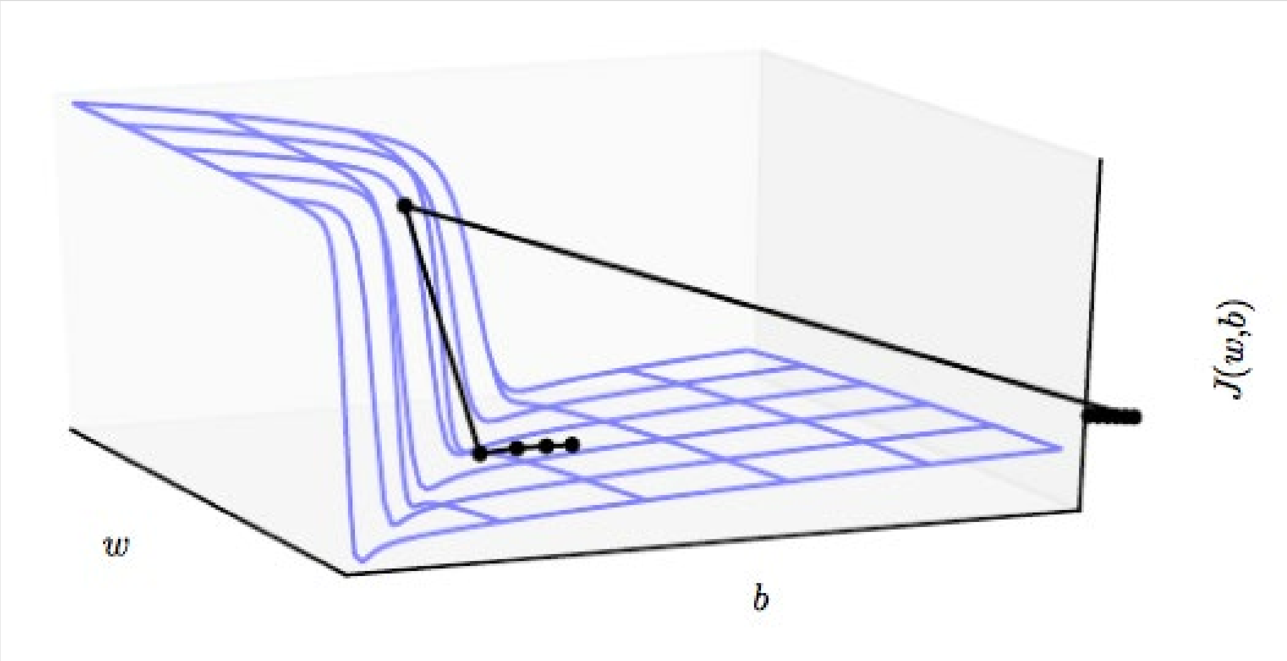
Exploding and Vanishing Gradients

Exploding gradients lead to cliffs

Can be mitigated using gradient clipping

$$\text{if } ||g|| > u$$

$$g \leftarrow \frac{gu}{||g||}$$



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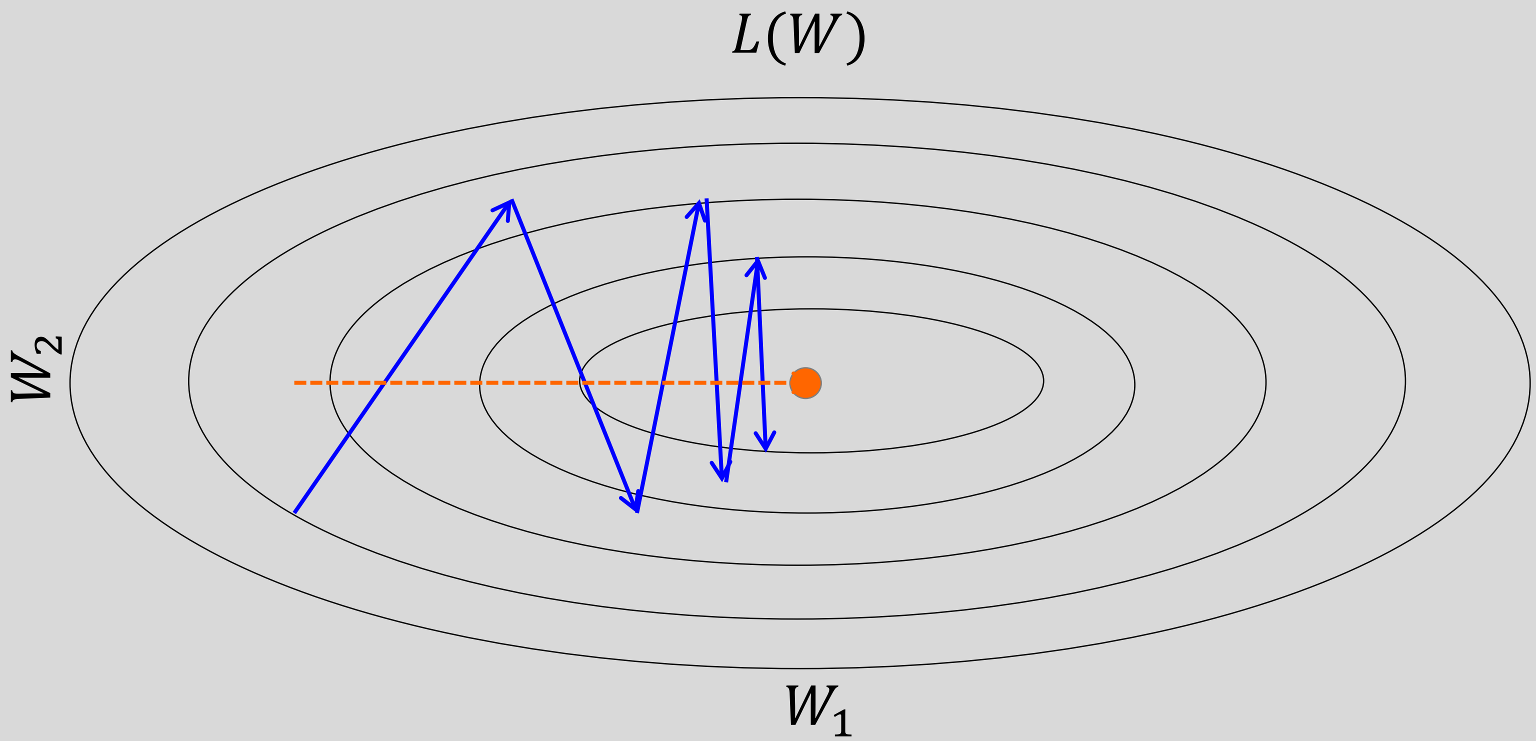
Momentum

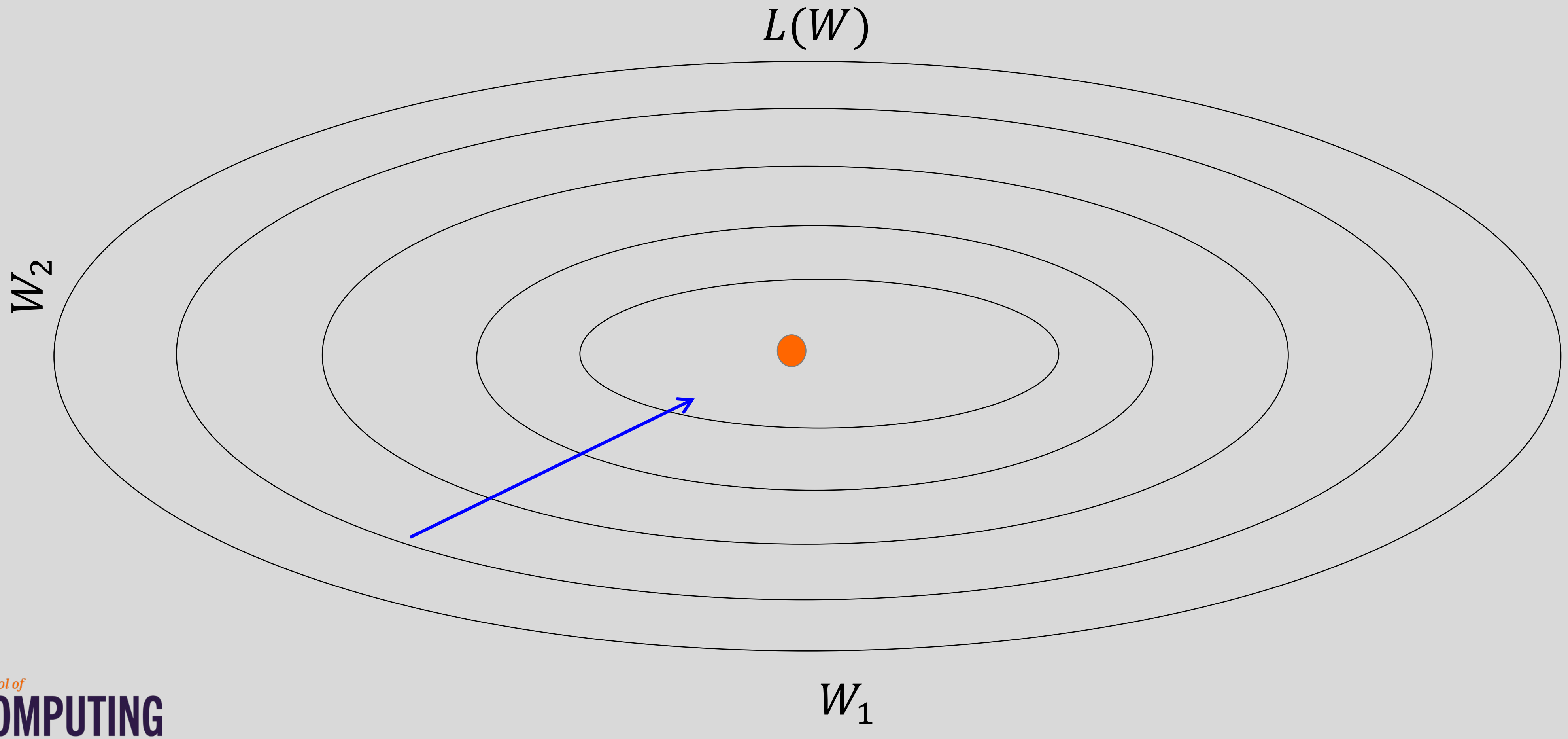
Oscillations because updates do not exploit curvature information



Average gradient presents faster path to optimal:
vertical components cancel out

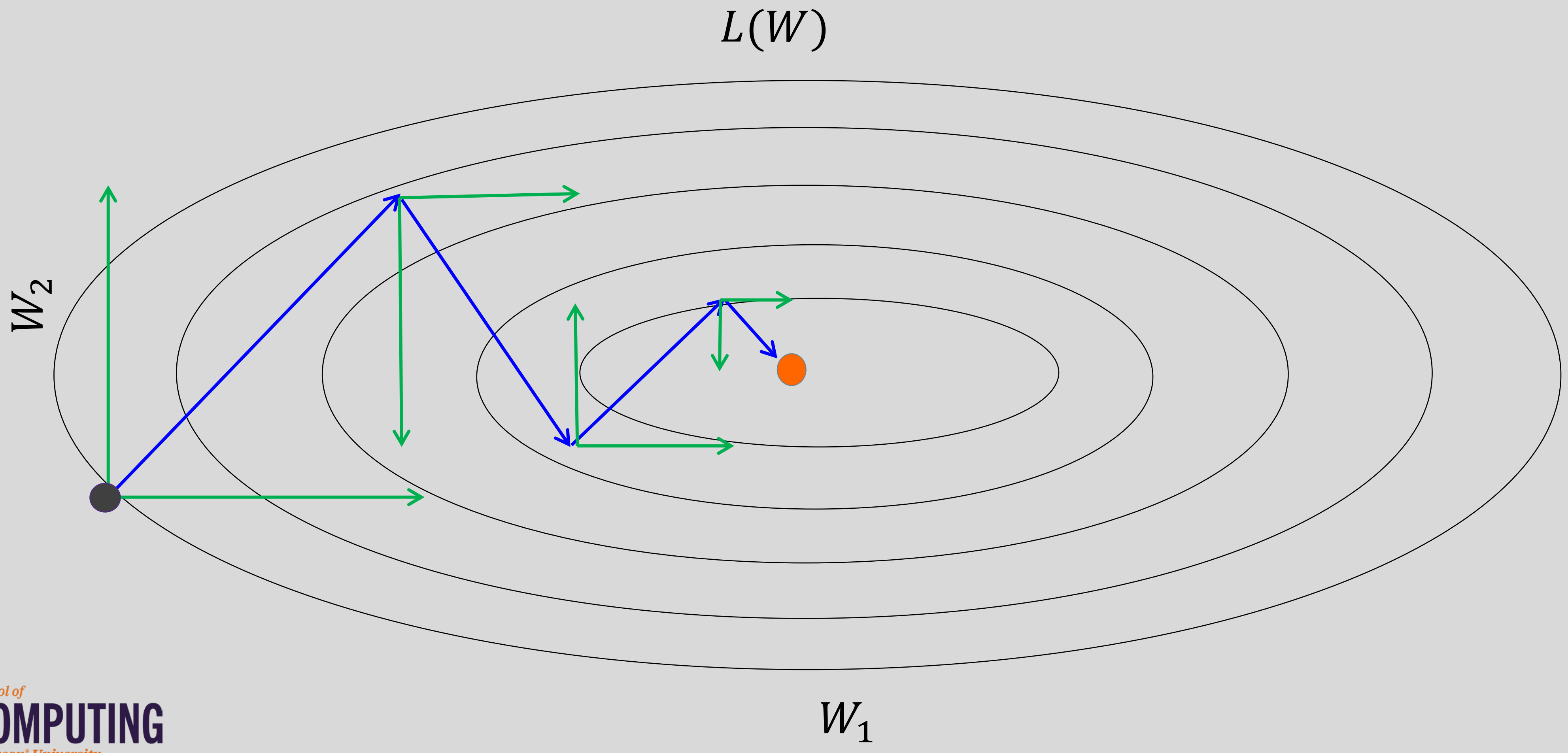
Momentum

Question: Why not this?



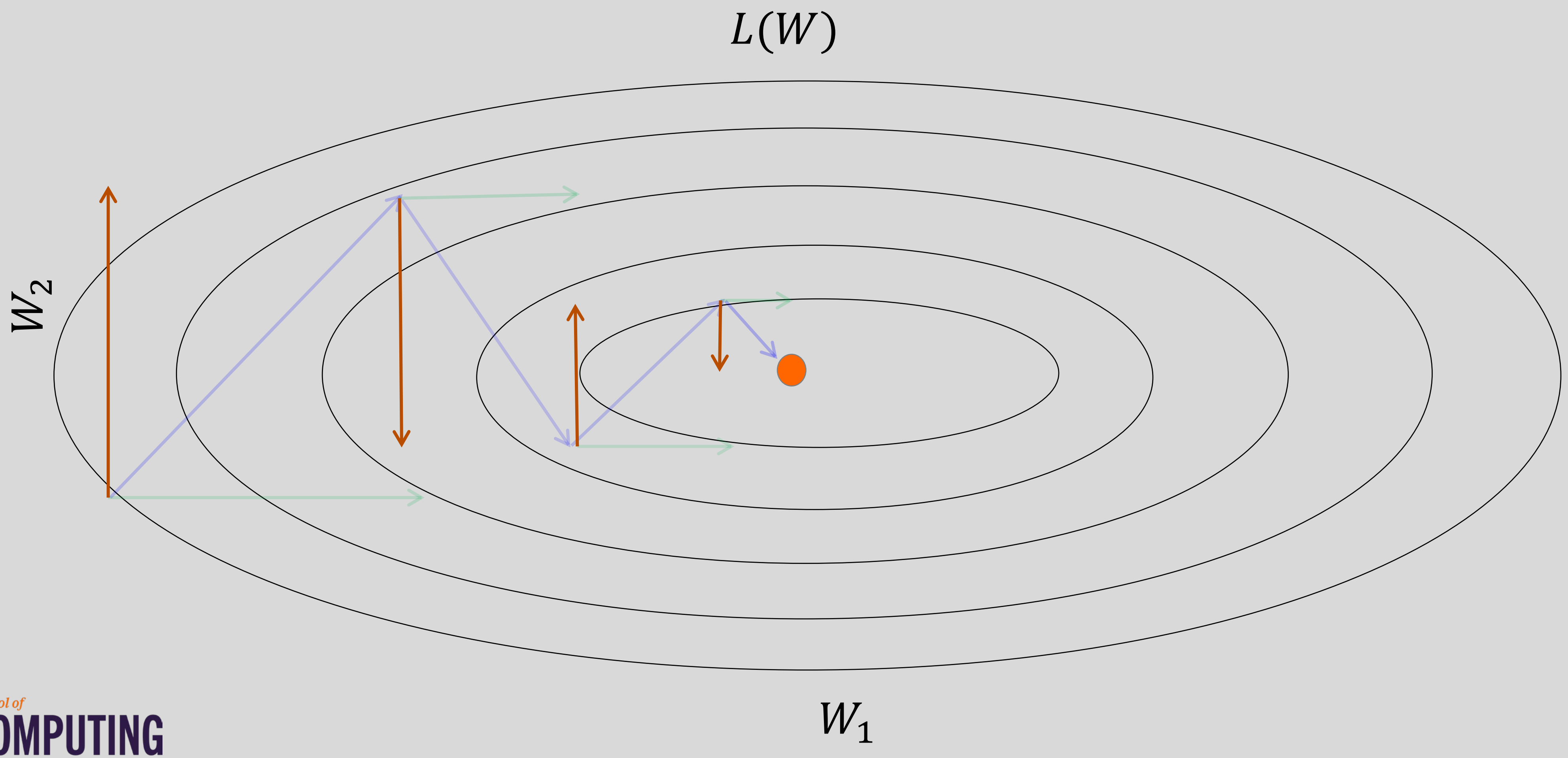
Momentum

Let us figure out an algorithm which will lead us to the minimum faster.



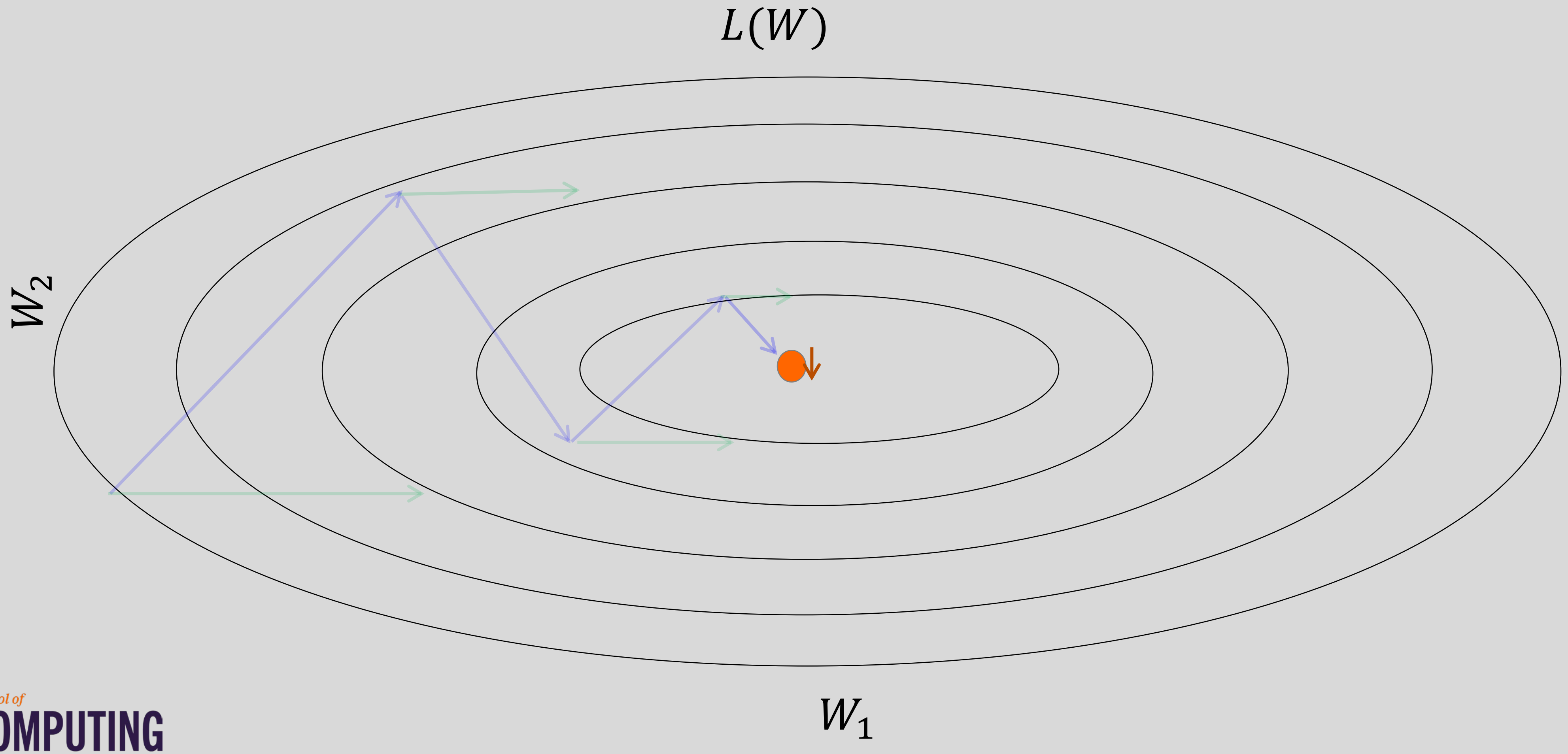
Momentum

Look each component at a time



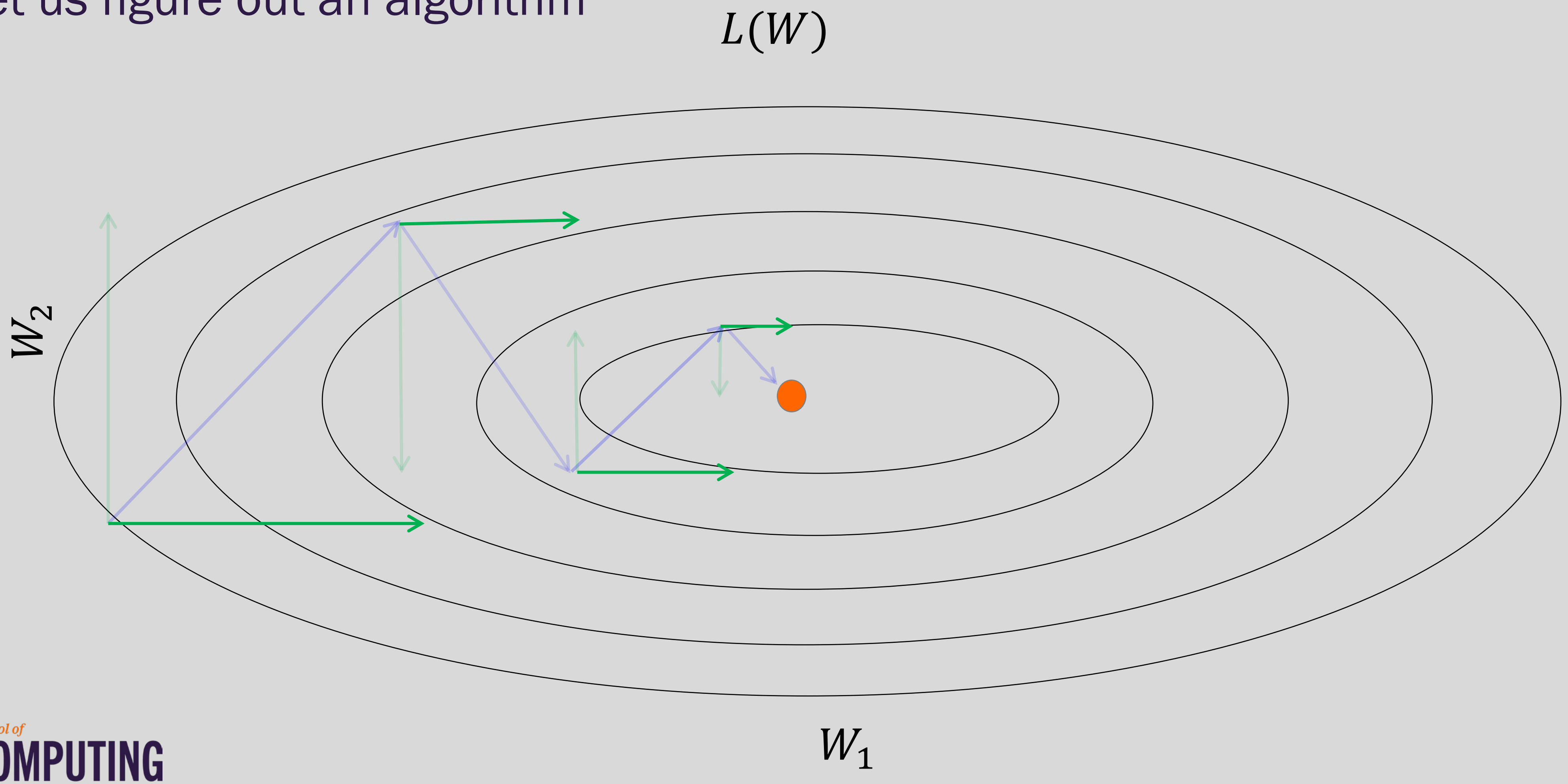
Momentum

Let us figure out an algorithm



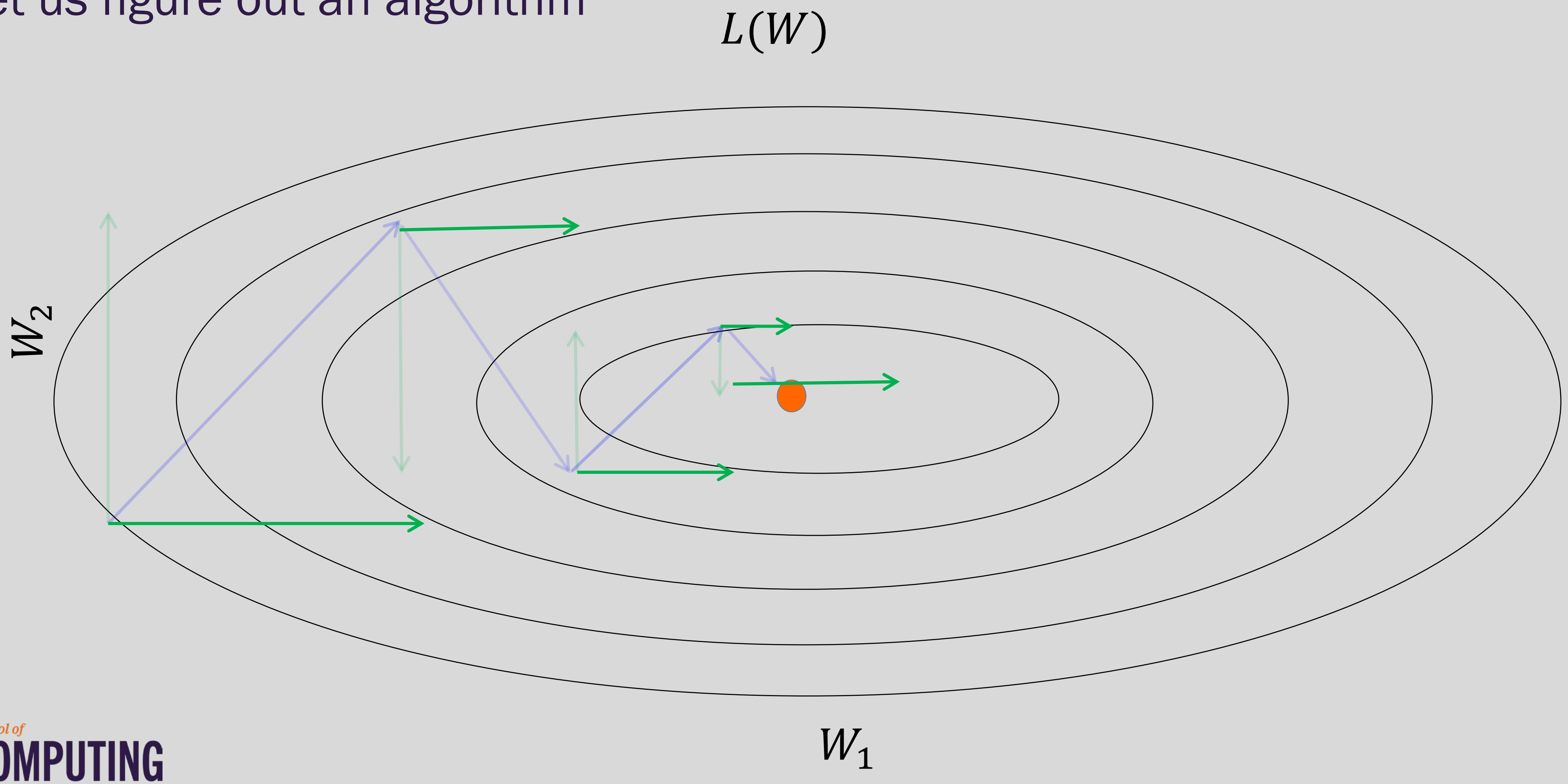
Momentum

Let us figure out an algorithm



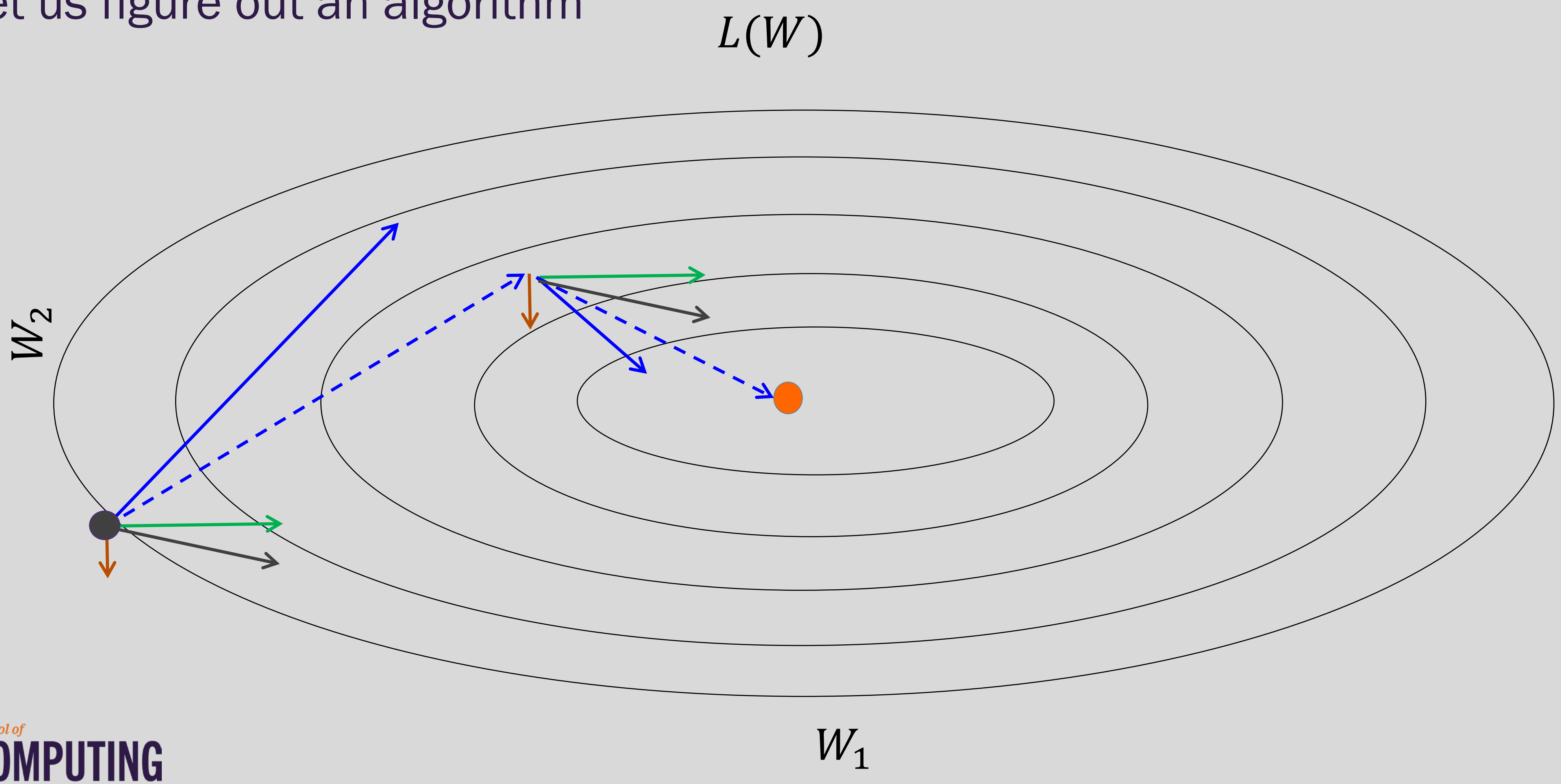
Momentum

Let us figure out an algorithm



Momentum

Let us figure out an algorithm



Momentum

f is the Neural Network

Old gradient descent:

$$g = \frac{1}{m}\sum_i \nabla_W L(f(x_i;W), y_i) \qquad W^* = W - \lambda g$$

$$g = g + average\ g\ from\ before$$

New gradient descent with momentum:

$$v = \alpha v + (1-\alpha)\, g \qquad W^* = W - \lambda v$$

$\alpha \in [0,1)$ controls how quickly effect of past gradients decay

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Nesterov Momentum

Apply an interim update: $\tilde{W} = W + v$

Perform a correction based on gradient at the interim point:

$$g = \frac{1}{m}\sum_i \nabla_W L(f(x_i;\tilde{W}), y_i)$$

$$v = \alpha v - \varepsilon g$$

$$W = W + v$$

Momentum based on look-ahead slope

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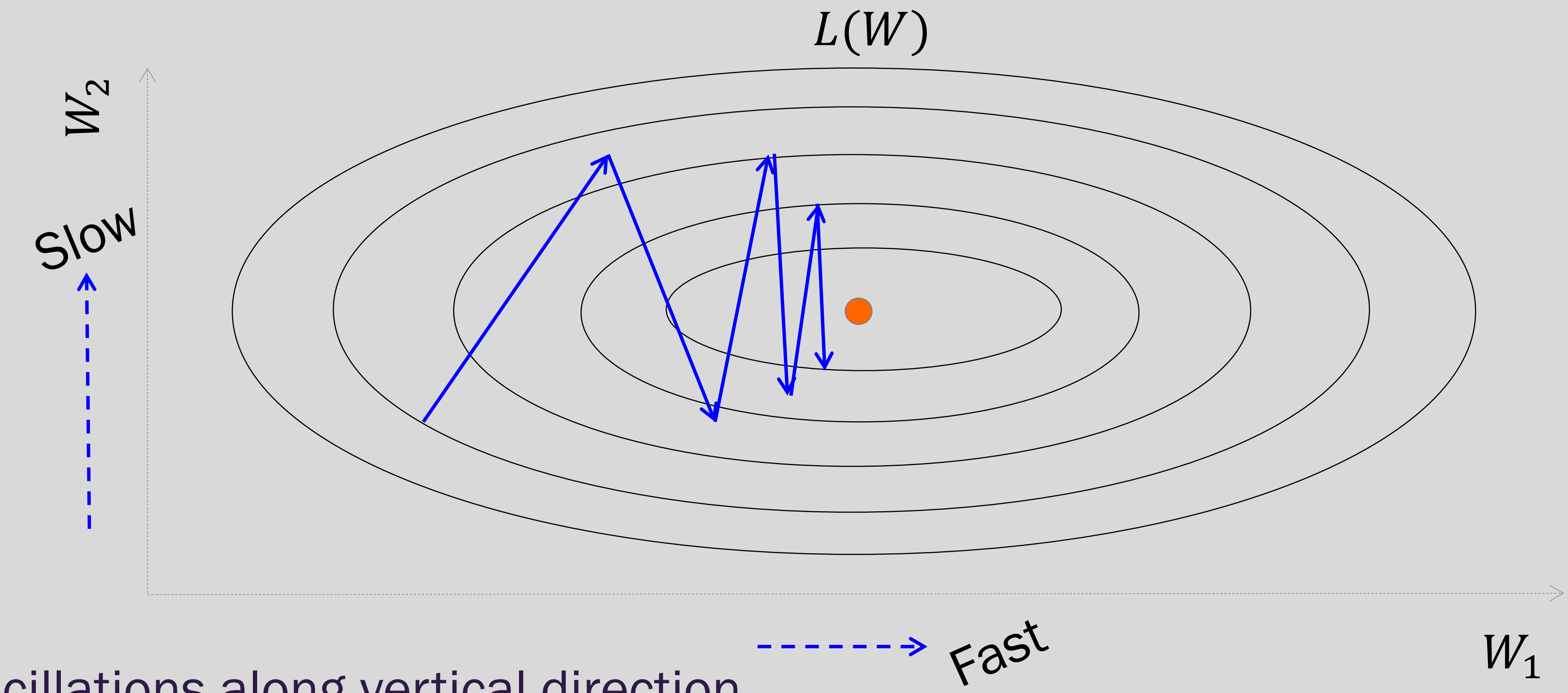
Adaptive Learning Rates

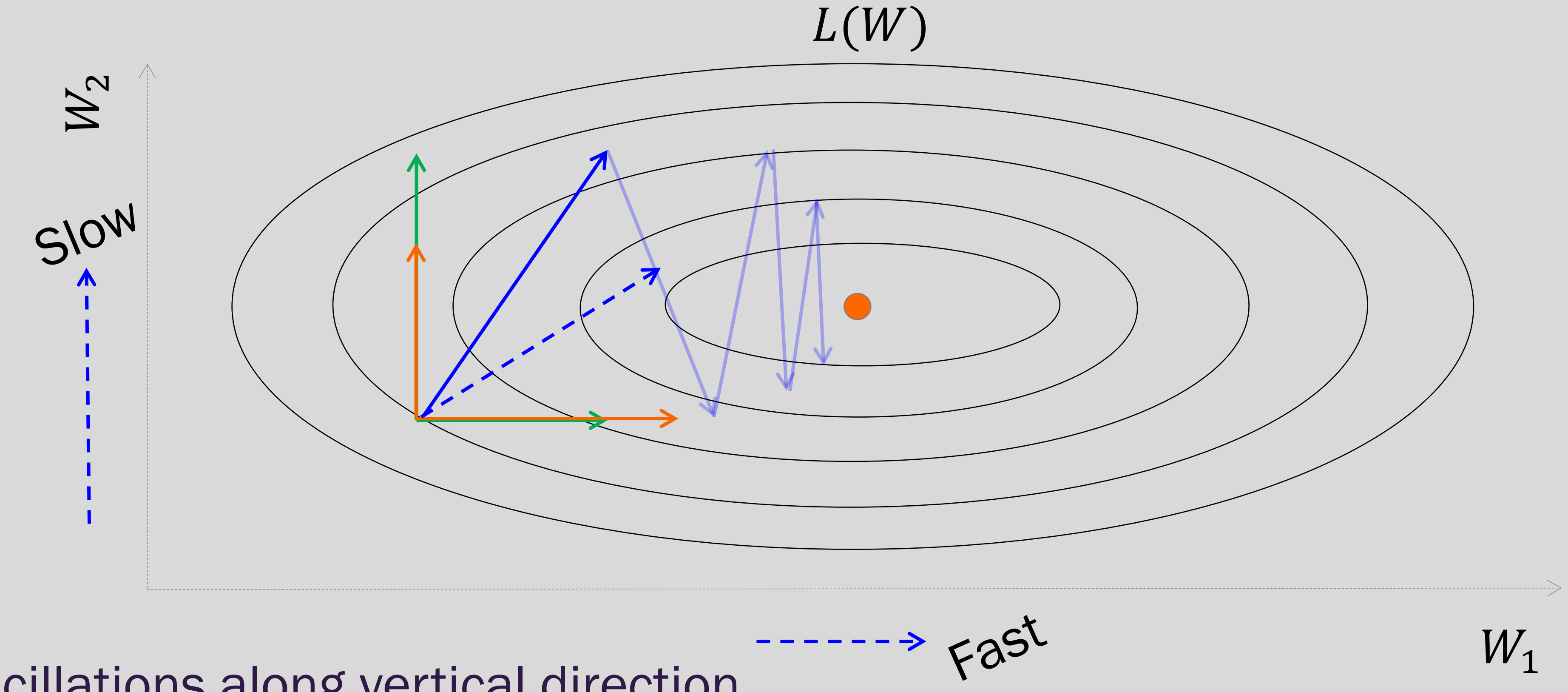


Oscillations along vertical direction

- Learning must be slower along parameter 2

Use a different learning rate for each parameter?

Adaptive Learning Rates



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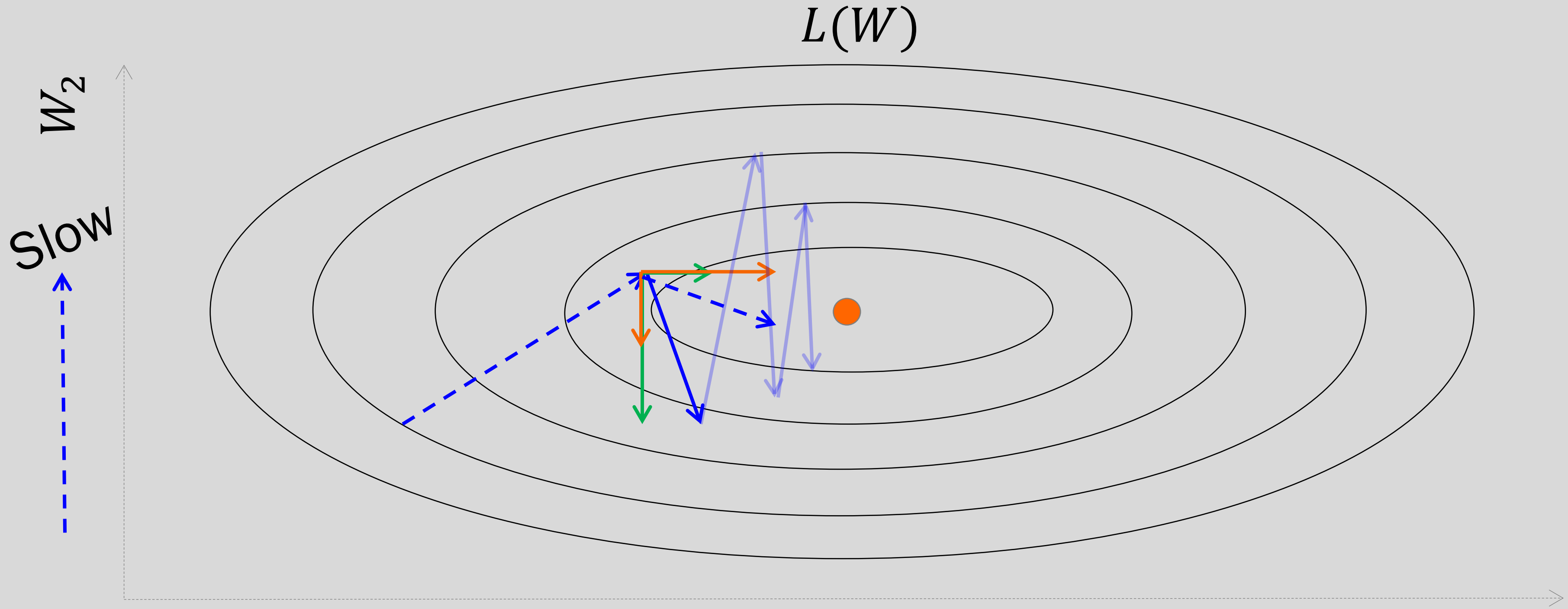

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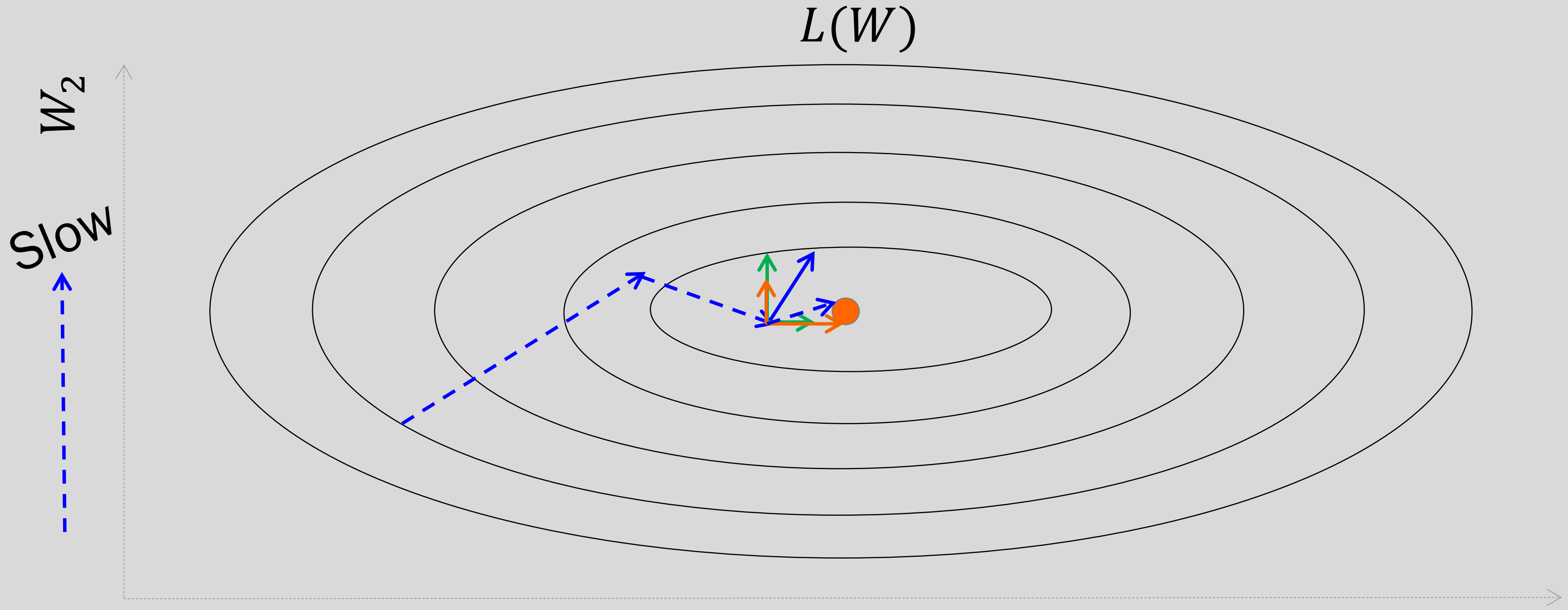

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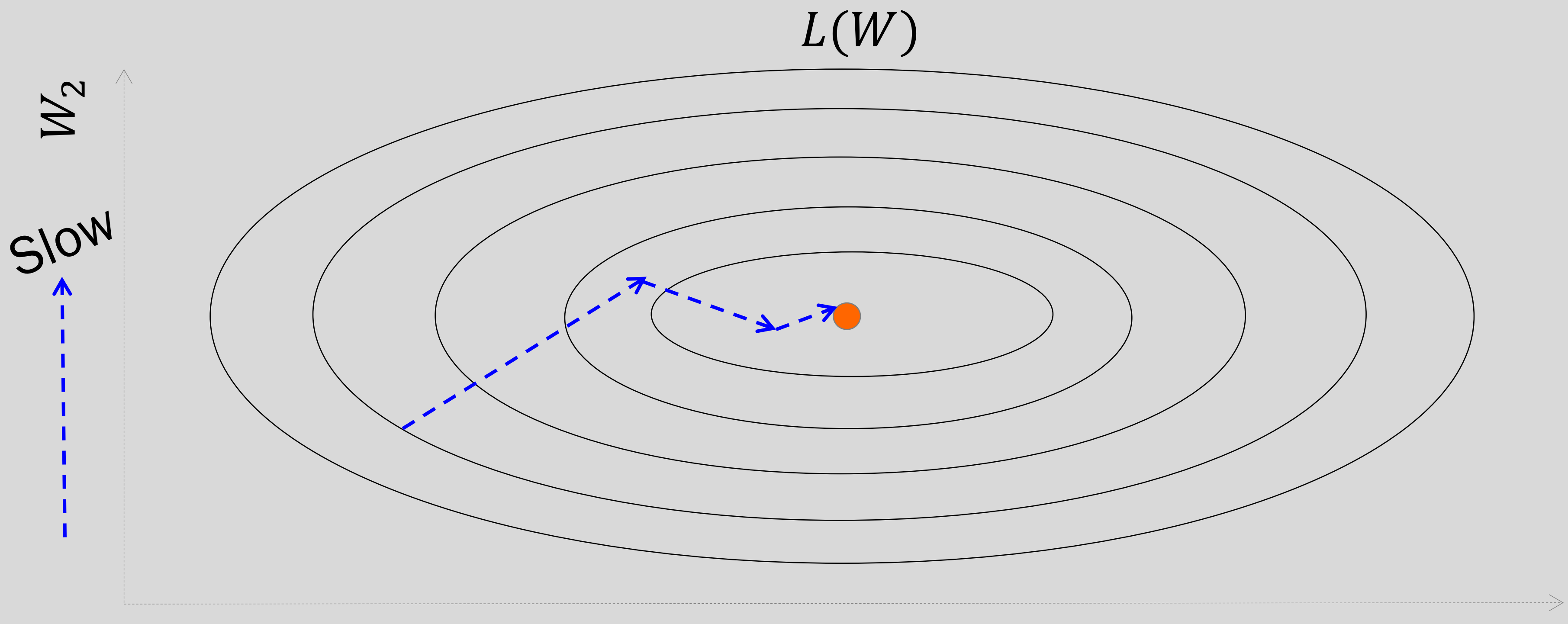

Oscillations along vertical direction

- Learning must be slower along parameter 2

Use a different learning rate for each parameter?

AdaGrad

- Accumulate squared gradients:

$$r_i = r_i + g_i^2$$

g is the gradient

- Update each parameter:

$$W_i = W_i - \frac{\epsilon}{\delta + \sqrt{r_i}} g_i$$

Inversely proportional to cumulative squared gradient

- Greater progress along gently sloped directions δ is a small number, making sure this does not become too large

AdaGrad

Old gradient descent:

$$g = \frac{1}{m}\sum_i \nabla_W L(f(x_i;W), y_i) \qquad W^* \quad W - \lambda g$$

We would like $\lambda's$ not to be the same and inversely proportional to the $|g_i|$

$$W_i^* = W_i - \lambda_i g_i \qquad \lambda_i \propto \frac{1}{|g_i|} = \frac{1}{\delta + |g_i|}$$

New gradient descent with adaptive learning rate:

$$r_i^* = r_i + g_i^2 \qquad W_i^* = W_i - \frac{\epsilon}{\delta + \sqrt{r_i}}\mathrm{g_i}$$

RMSProp

- For non-convex problems, AdaGrad can prematurely decrease learning rate
- Use exponentially weighted average for gradient accumulation

$$r_i = \rho r_i + (1-\rho) g_i^2$$

$$W_i = W_i - \frac{\epsilon}{\delta + \sqrt{r_i}} g_i$$

Adam

- RMSProp + Momentum
- Estimate first moment:

$$v_i = \rho_1 v_i + (1-\rho_1) g_i$$

Also applies bias correction to v and r

- Estimate second moment:

$$r_i = \rho_2 r_i + (1-\rho_2) g_i^2$$

- Update parameters:

$$W_i = W_i - \frac{\epsilon}{\delta + \sqrt{r_i}} \nu_i$$

Works well in practice, is fairly robust to hyper-parameters

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Parameter Initialization

- Goal: break symmetry between units
  - so that each unit computes a different function
- Initialize all weights (not biases) randomly
  - Gaussian or uniform distribution
- Scale of initialization?
  - *Large* -> grad explosion, *Small* -> grad vanishing

Xavier Initialization

- Heuristic for all outputs to have unit variance
- For a fully-connected layer with m inputs:

$$W_{ij} \sim N\left(0,\ \frac{1}{m}\right)$$

- For ReLU units, it is recommended:

$$W_{ij} \sim N\left(0,\ \frac{2}{m}\right)$$

Normalized Initialization

- Fully-connected layer with m inputs, n outputs:

$$W_{ij} \sim U\left(-\sqrt{\frac{6}{m+n}}, \sqrt{\frac{6}{m+n}}\right)$$

- Heuristic trades off between initialize all layers have same activation and gradient variance
- Sparse variant when m is large
  - Initialize k nonzero weights in each unit

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Feature Normalization

Good practice to normalize features before applying learning algorithm:

Feature vector

Vector of mean feature values

$$x' = \frac{x - \mu}{\sigma}$$

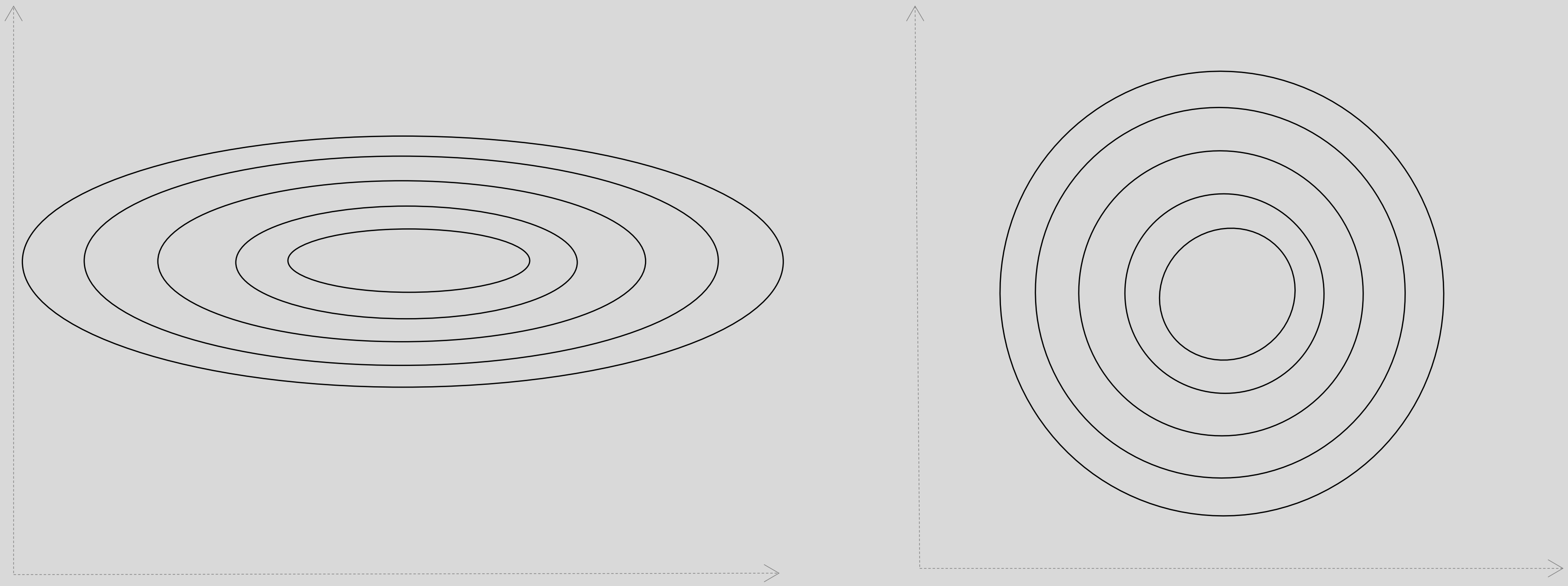
Vector of SD of feature values

Features in same scale: mean 0 and variance 1

- Speeds up learning

Feature Normalization

$L(W)$

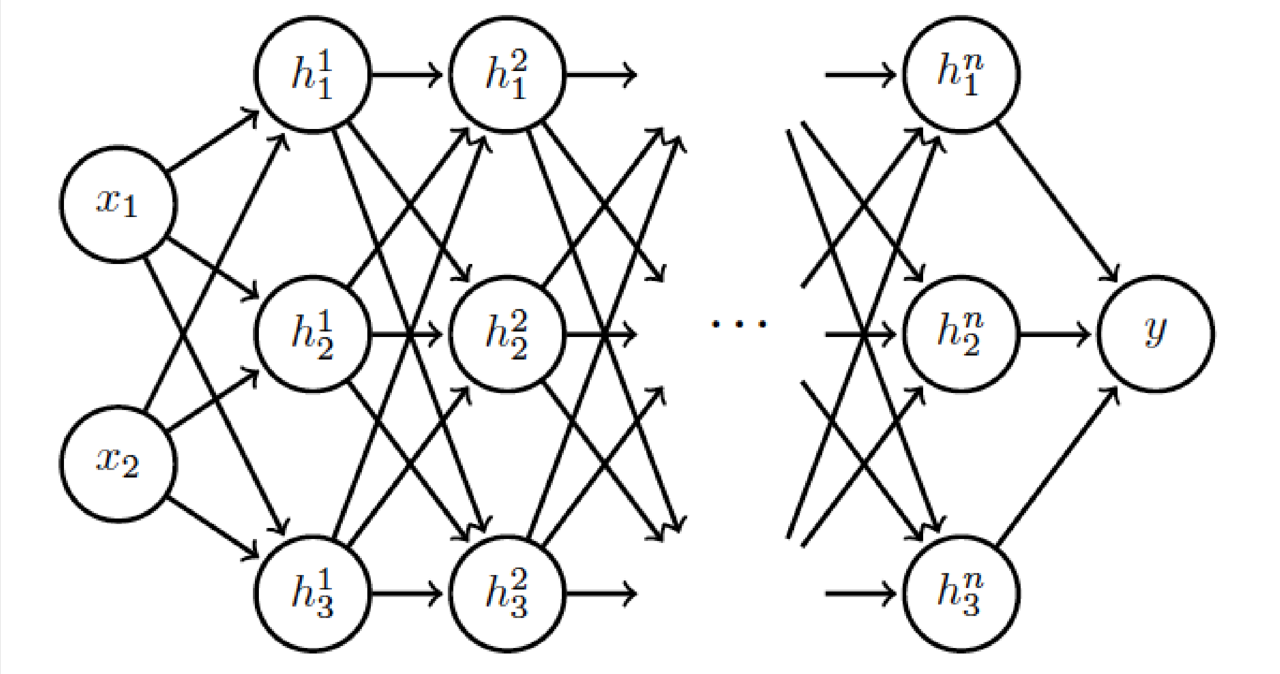


Before normalization

After normalization

Internal Covariance Shift

Each hidden layer changes distribution of inputs to next layer: *slows down learning*



Normalize inputs to layer 2

…

Normalize inputs to layer n

Batch Normalization

Training time:

- Mini-batch of activations for layer to normalize

$$H = \begin{bmatrix} H_{11} & \cdots & H_{1K} \\ \vdots & \ddots & \vdots \\ H_{N1} & \cdots & H_{NK} \end{bmatrix}$$

K hidden layer activations

N data points in mini-batch

Batch Normalization

Training time:

- Mini-batch of activations for layer to normalize

$$H' = \frac{H - \mu}{\sigma}$$

where

$$\mu = \frac{1}{m}\sum_i H_{i,:} \qquad \sigma = \sqrt{\frac{1}{m}\sum_i (H - \mu)_i^2 + \delta}$$

Vector of mean activations across mini-batch

Vector of SD of each unit across mini-batch

Batch Normalization

Training time:

- Normalization can reduce expressive power
- Instead use:

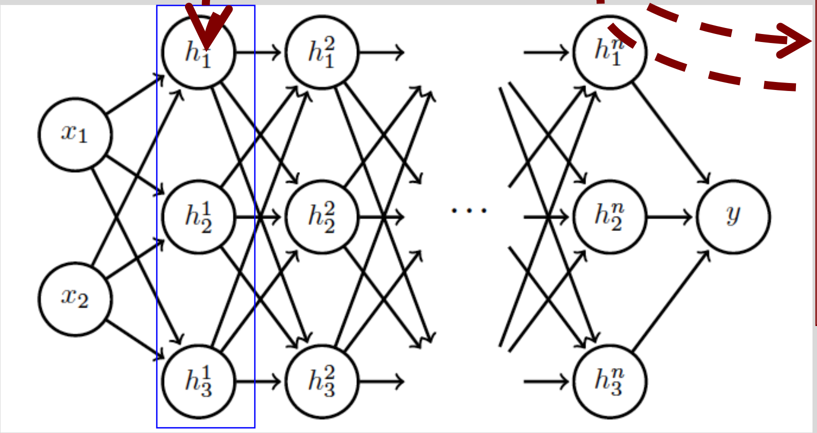
$$\gamma H' + \beta$$

Learnable parameters

- Allows network to control range of normalization

Batch Normalization

Batch 1

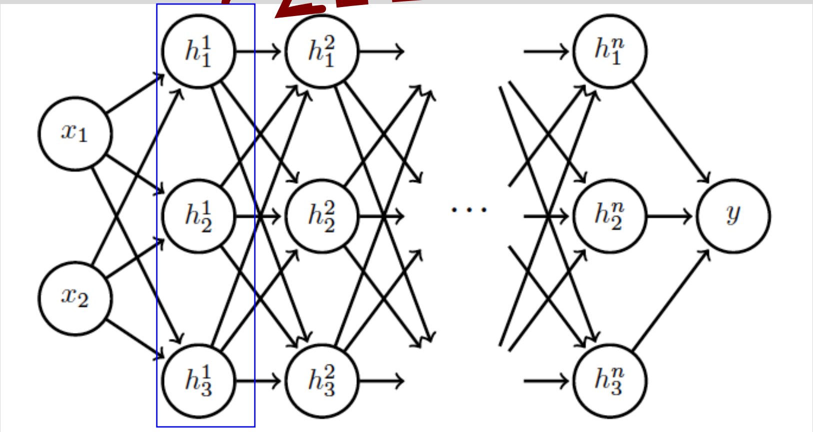


$$\mu^1 = \frac{1}{m} \sum_i H_{i,:}$$

$$\sigma^1 = \sqrt{\frac{1}{m} \sum_i (H - \mu)_i^2 + \delta}$$

⋮

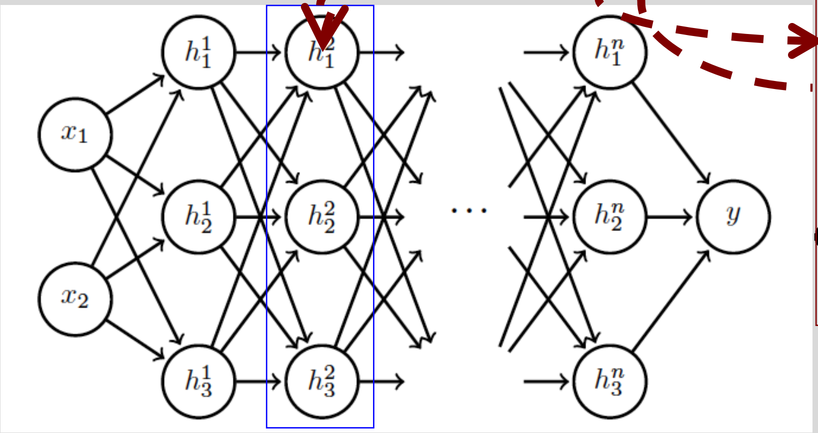
Batch *N*



Add normalization operations for layer 1

Batch Normalization

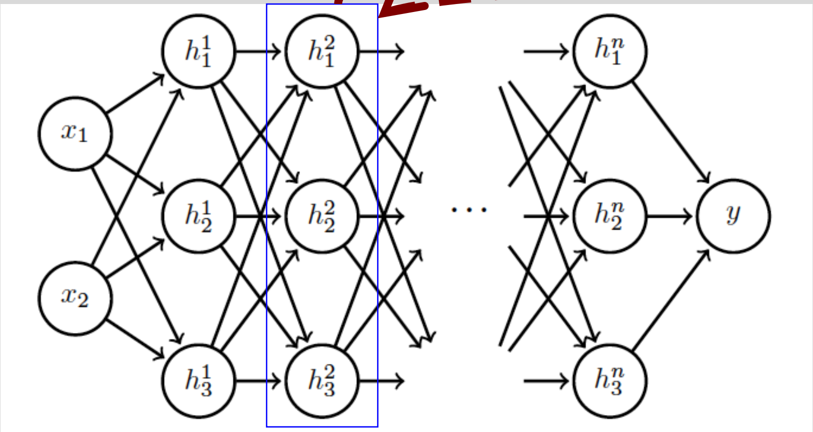
Batch 1



$$\mu^2 = \frac{1}{m}\sum_i H_{i,:}$$

$$\sigma^2 = \sqrt{\frac{1}{m}\sum_i (H-\mu)_i^2 + \delta}$$

Batch *N*



Add normalization operations for layer 2 and so on ...

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Regularization

Regularization is any modification we make to a learning algorithm that is intended to **reduce its generalization error** but not its training error.

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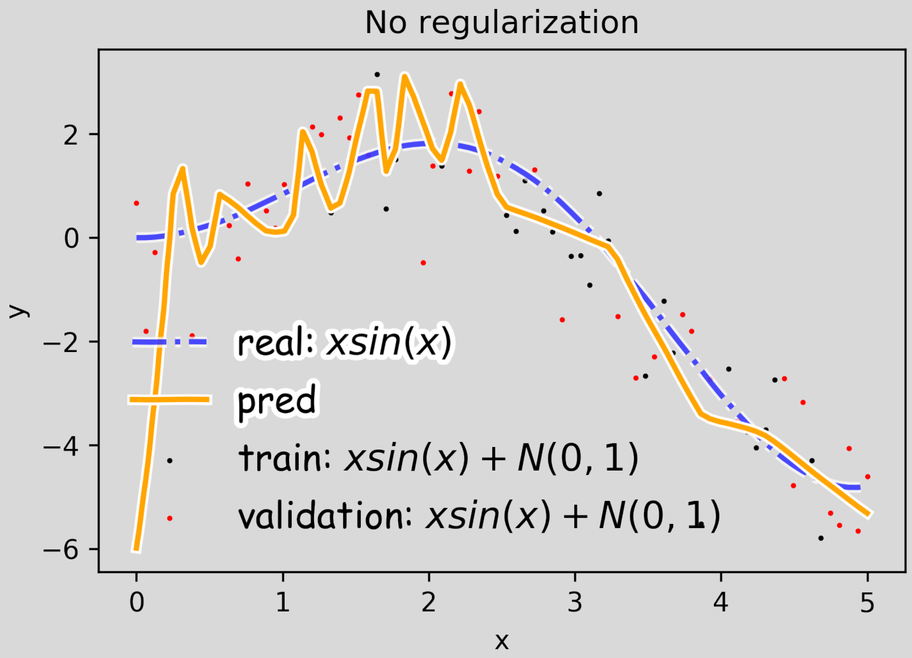
Overfitting

Fitting a deep neural network with 5 layers and 100 neurons per layer can lead to a very good prediction on the training set but poor prediction on validations set.



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Norm Penalties

We used to optimize:

$$L(W; X, y)$$

Change to …

$$L_R(W; X, y) = L(W; X, y) + \alpha\Omega(W)$$

Biases not penalized

L_2 regularization:

- Weights decay
- MAP estimation with Gaussian prior

$$\Omega(W) = \frac{1}{2} \| W \|_2^2$$

L_1 regularization:

- encourages sparsity
- MAP estimation with Laplacian prior

$$\Omega(W) = \frac{1}{2} \| W \|_1$$

Norm Penalties

We used to op[...]

$$W^{(i+1)} = W^{(i)} - \lambda \frac{\partial L}{\partial W}$$

Change to ...

$$L_R(W; X, y) = L(W; X, y) + \frac{1}{2}\alpha W^2$$

$$W^{(i+1)} = W^{(i)} - \lambda \frac{\partial L}{\partial W} \boxed{- \lambda\alpha\, W}$$

Weights decay in proportion to size

Biases not penalized

L_2 regularization:

- **Decay of weights**
- MAP estimation with Gaussian prior

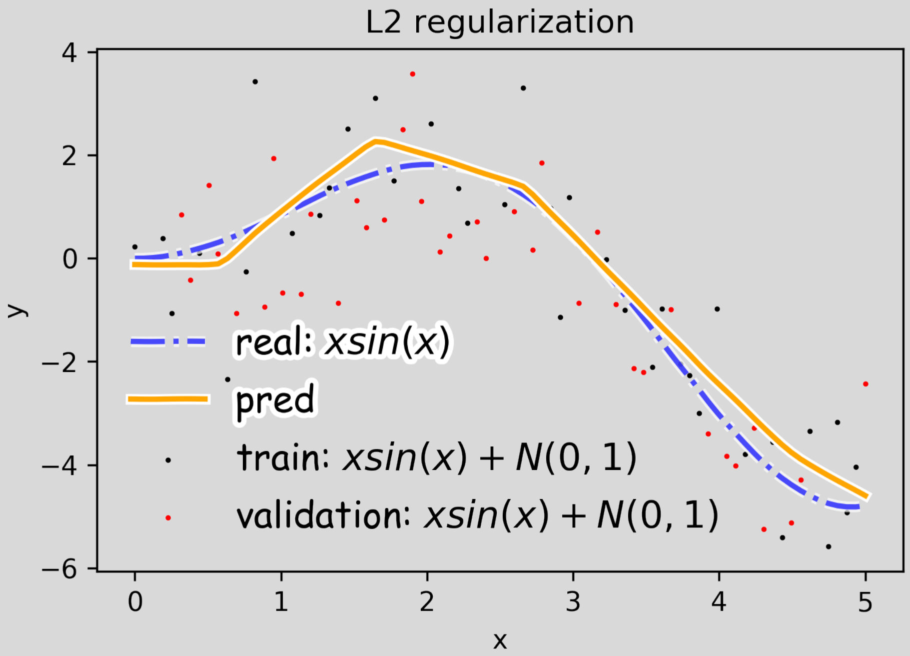
$$\Omega(W) = \frac{1}{2} \parallel W \parallel_2^2$$

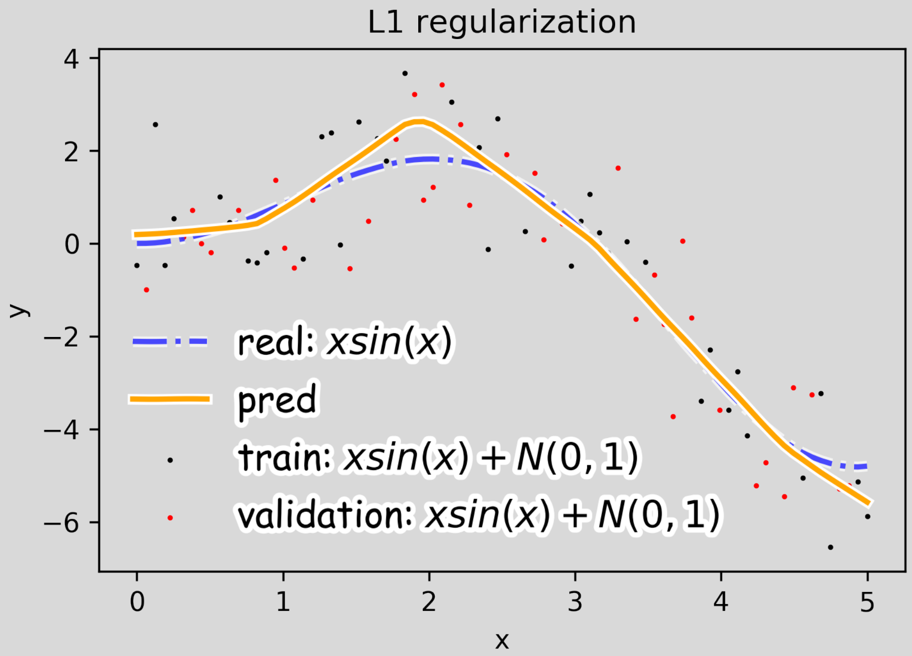
L_1 regularization:

- encourages sparsity
- MAP estimation with Laplacian prior

$$\Omega(W) = \frac{1}{2} \parallel W \parallel_1$$

Norm Penalties





$$\Omega(W) = \frac{1}{2} \parallel W \parallel_2^2$$

$$\Omega(\mathrm{W}) = \frac{1}{2} \parallel \mathrm{W} \parallel_1$$

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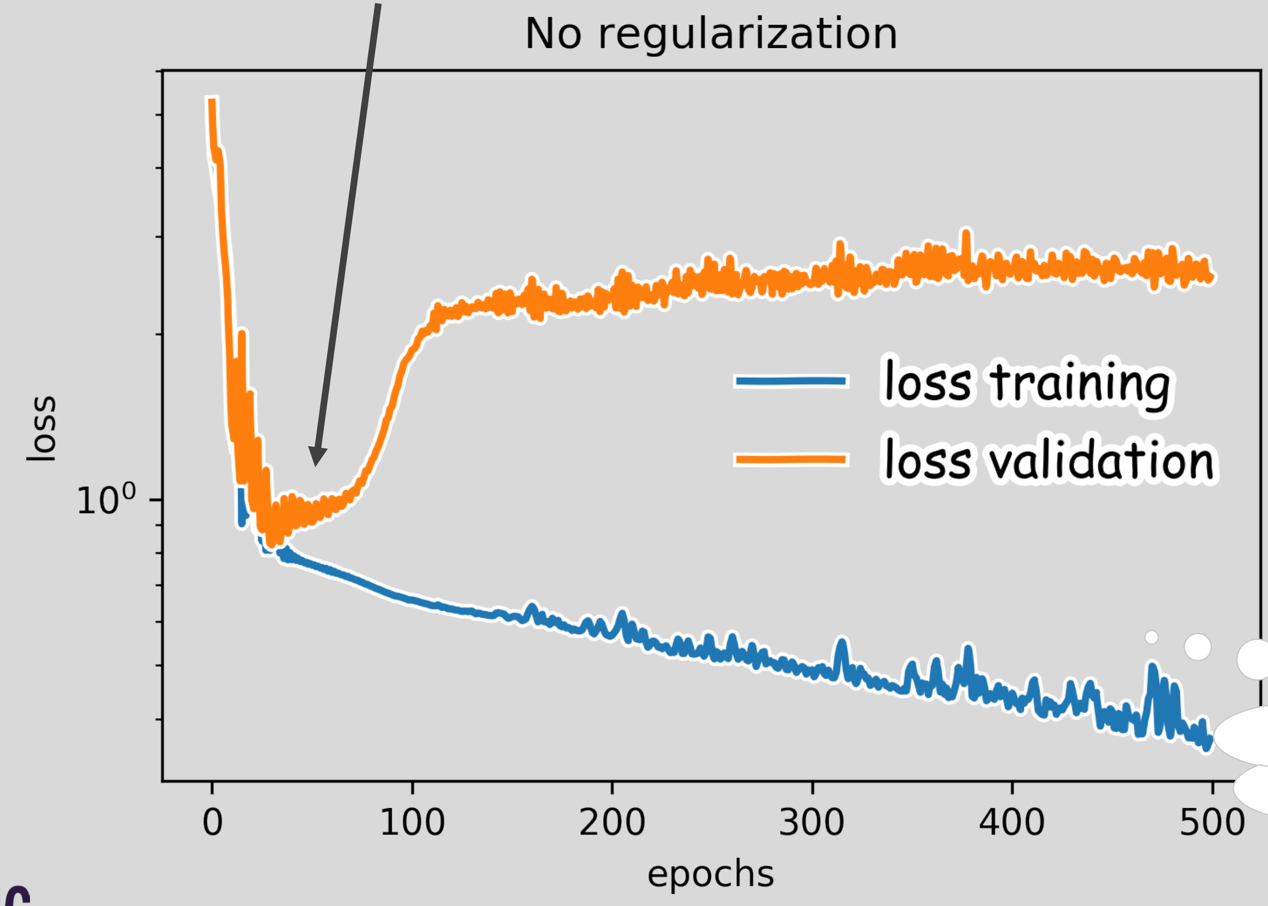
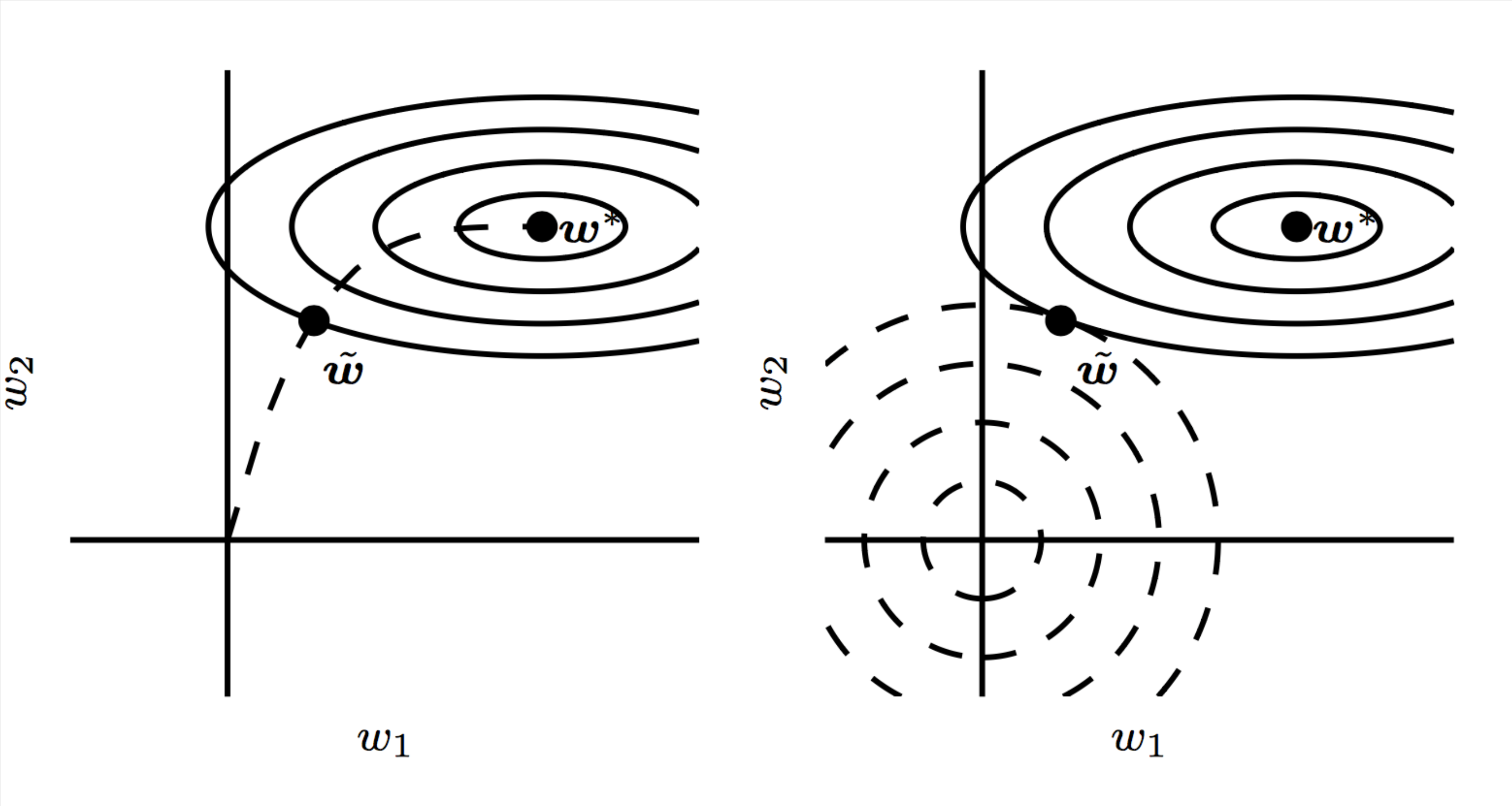
Early Stopping

Early stopping: terminate while validation set performance is better



Training time can be treated as a hyperparameter

Early Stopping



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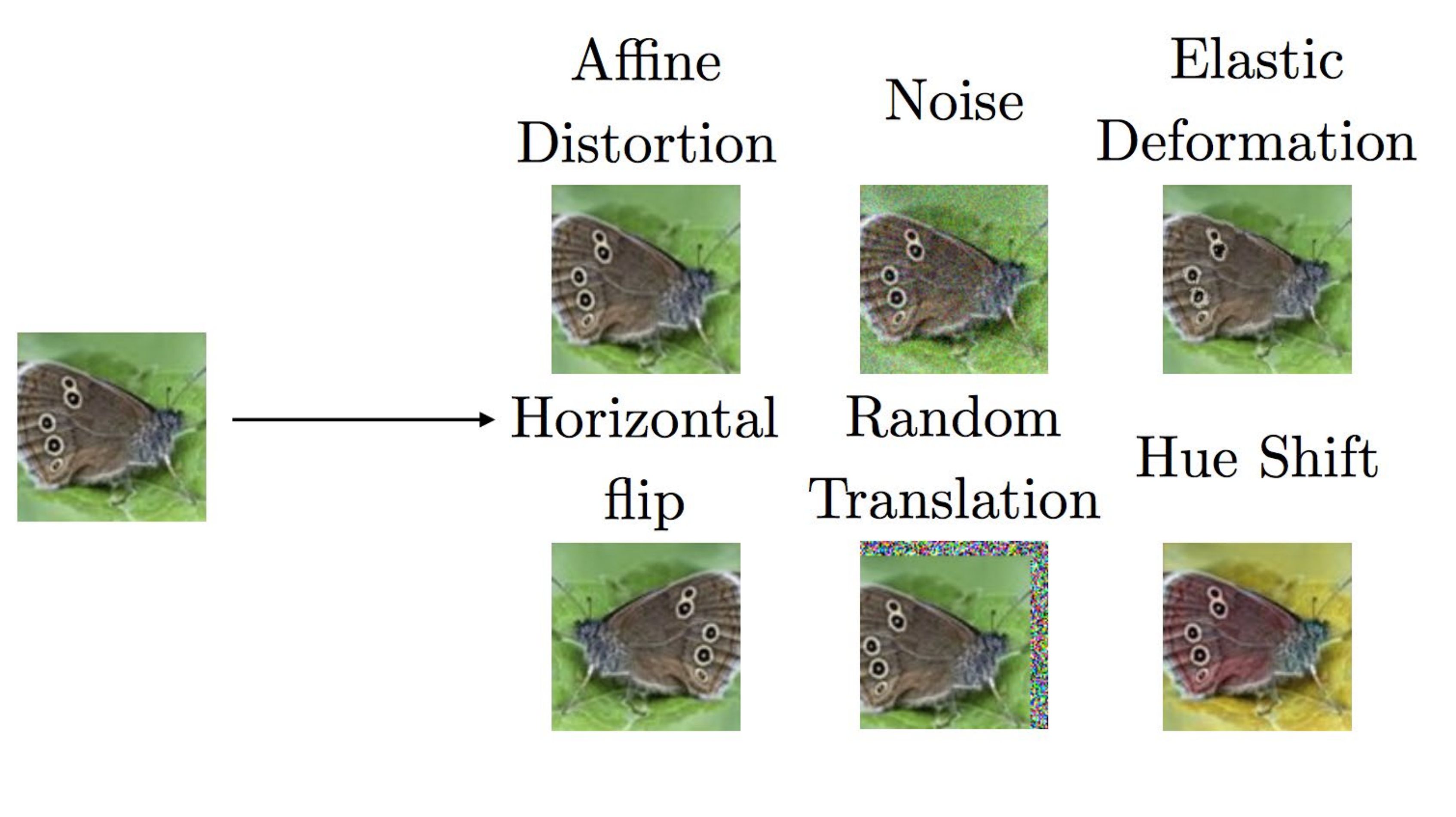
Data Augmentation



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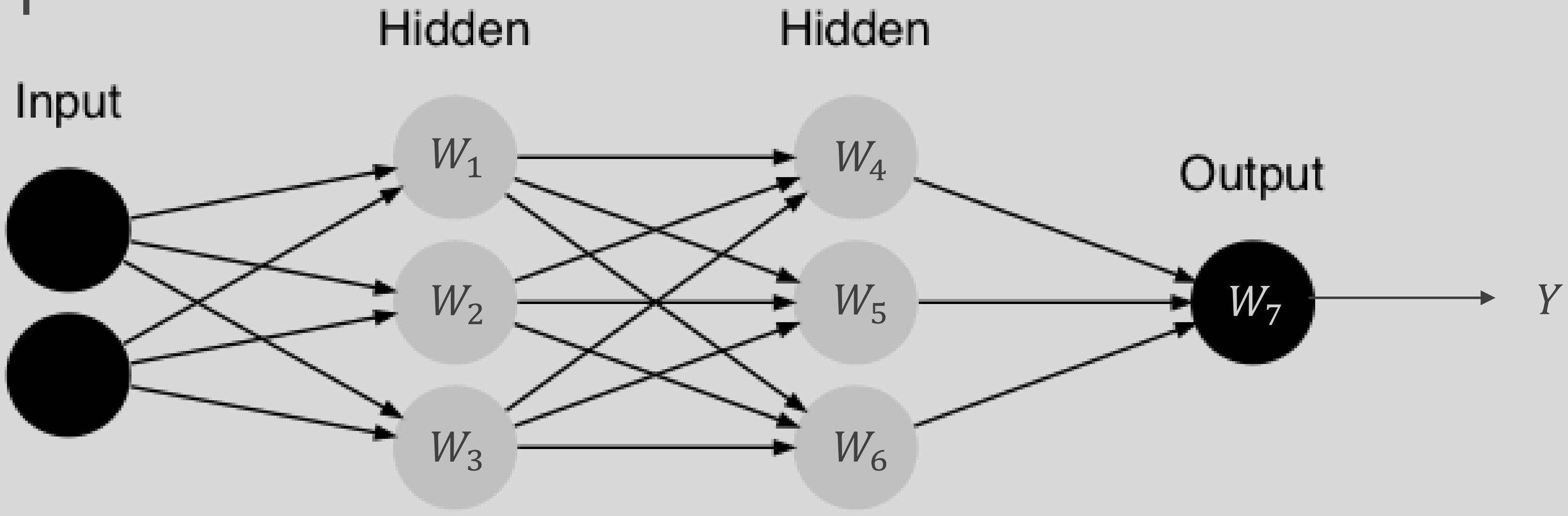
Sparse Representation



$$L(\theta; X, y)$$

$$[4.34] = \underbrace{[3.2 \quad 2.0 \quad 1.8]}_{W_7} \begin{bmatrix} 2 \\ -2.2 \\ 1.3 \end{bmatrix}$$

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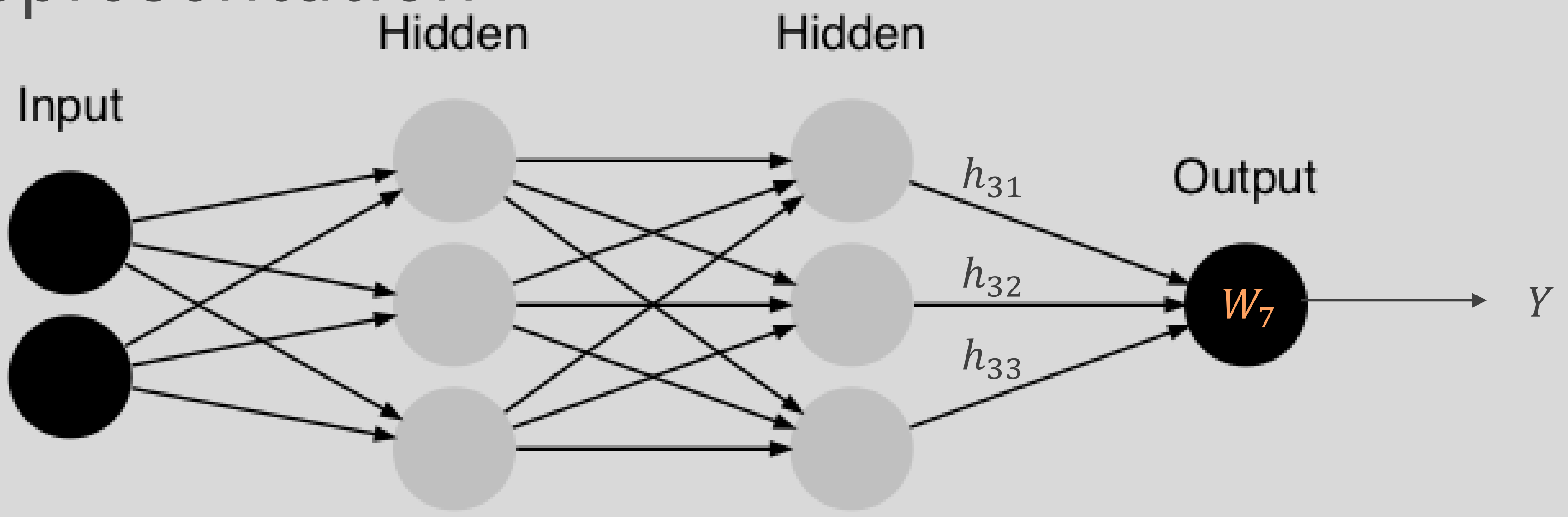
Sparse Representation



$$L(\theta; X, y)$$

$$[4.34] = [3.2 \quad 2 \quad 1] \underbrace{\begin{bmatrix} 2 \\ -2.2 \\ 1.3 \end{bmatrix}}_{h_{31}, h_{32}, h_{33}}$$

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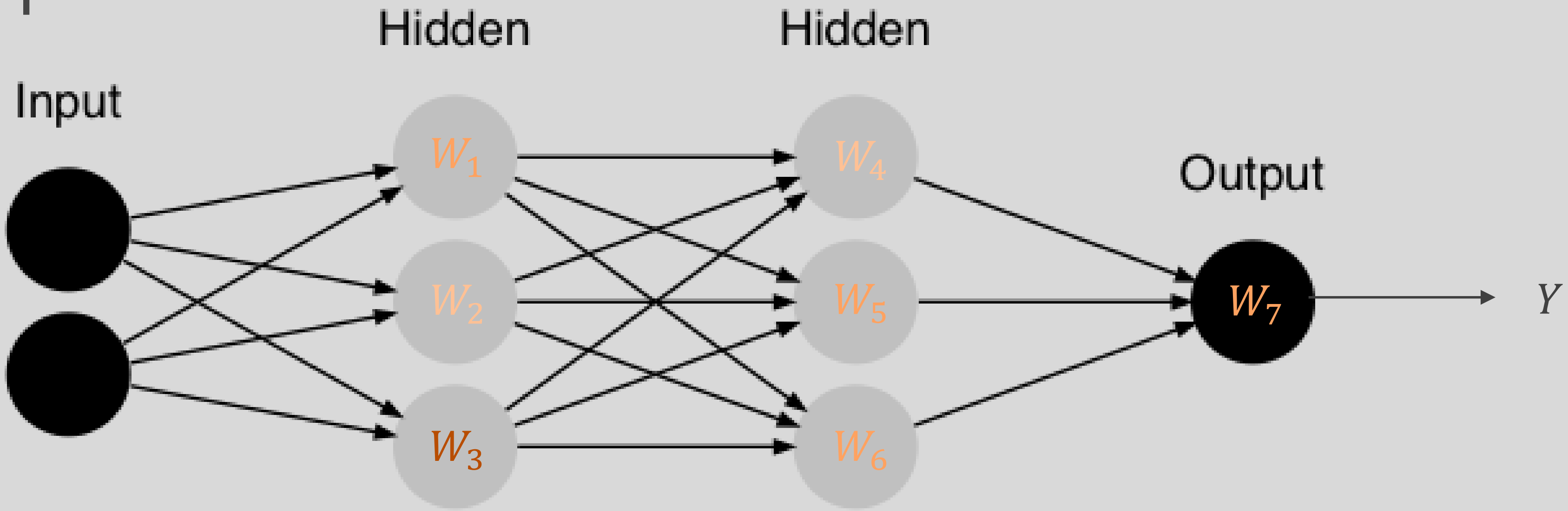
Sparse Representation

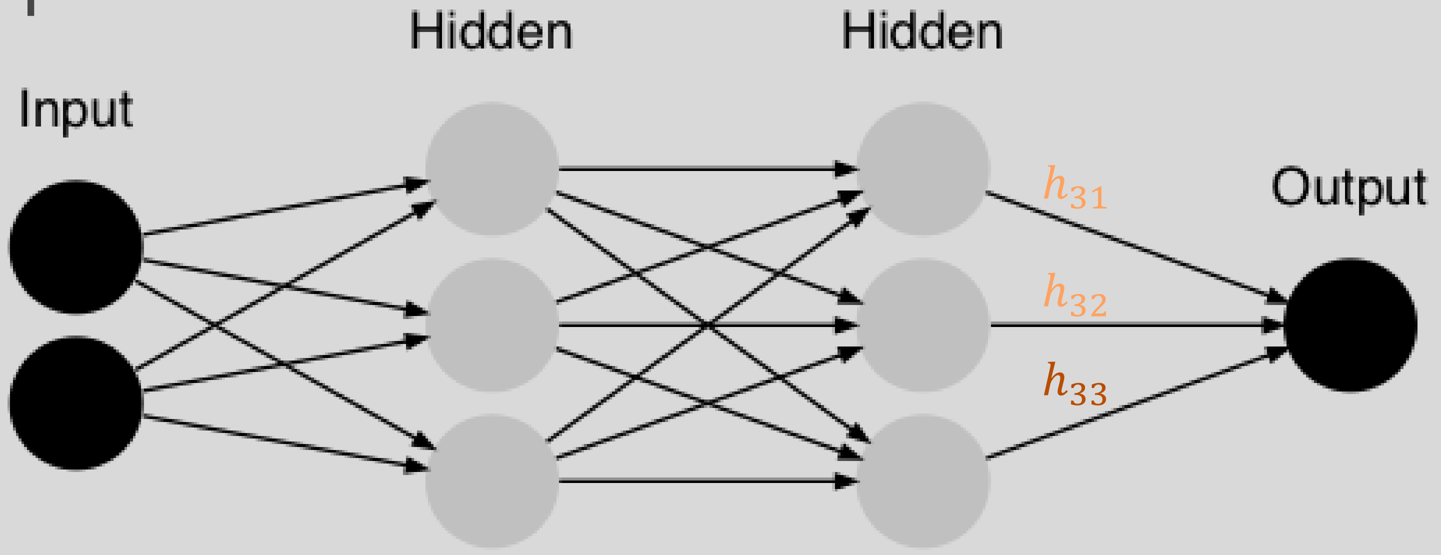


$$L_R(W; X, y) = L(\theta; X, y) + \alpha\Omega(W)$$

$$[0.69] = \underbrace{[0.5 \quad .2 \quad 0.1]}_{W_7} \begin{bmatrix} 2 \\ -2.2 \\ 1.3 \end{bmatrix}$$

Weights in output layer

Sparse Representation



$$L_R(W;X,y) = L(\theta;X,y) + \alpha\Omega(h)$$

$$[1.3] = [3.2 \quad 2 \quad 1]\begin{bmatrix} 0 \\ -0.2 \\ .9 \end{bmatrix} \quad \left.\vphantom{\begin{bmatrix} 0 \\ -0.2 \\ .9 \end{bmatrix}}\right\} h_{31}, h_{32}, h_{33}$$

Output of hidden layer

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Outline

Optimization

- Challenges in Optimization
- Momentum
- Adaptive Learning Rate
- Parameter Initialization
- Batch Normalization

Regularization of NN

- Norm Penalties
- Early Stopping
- Data Augmentation
- Sparse Representation
- **Dropout**

Noise Robustness

Random perturbation of network weights

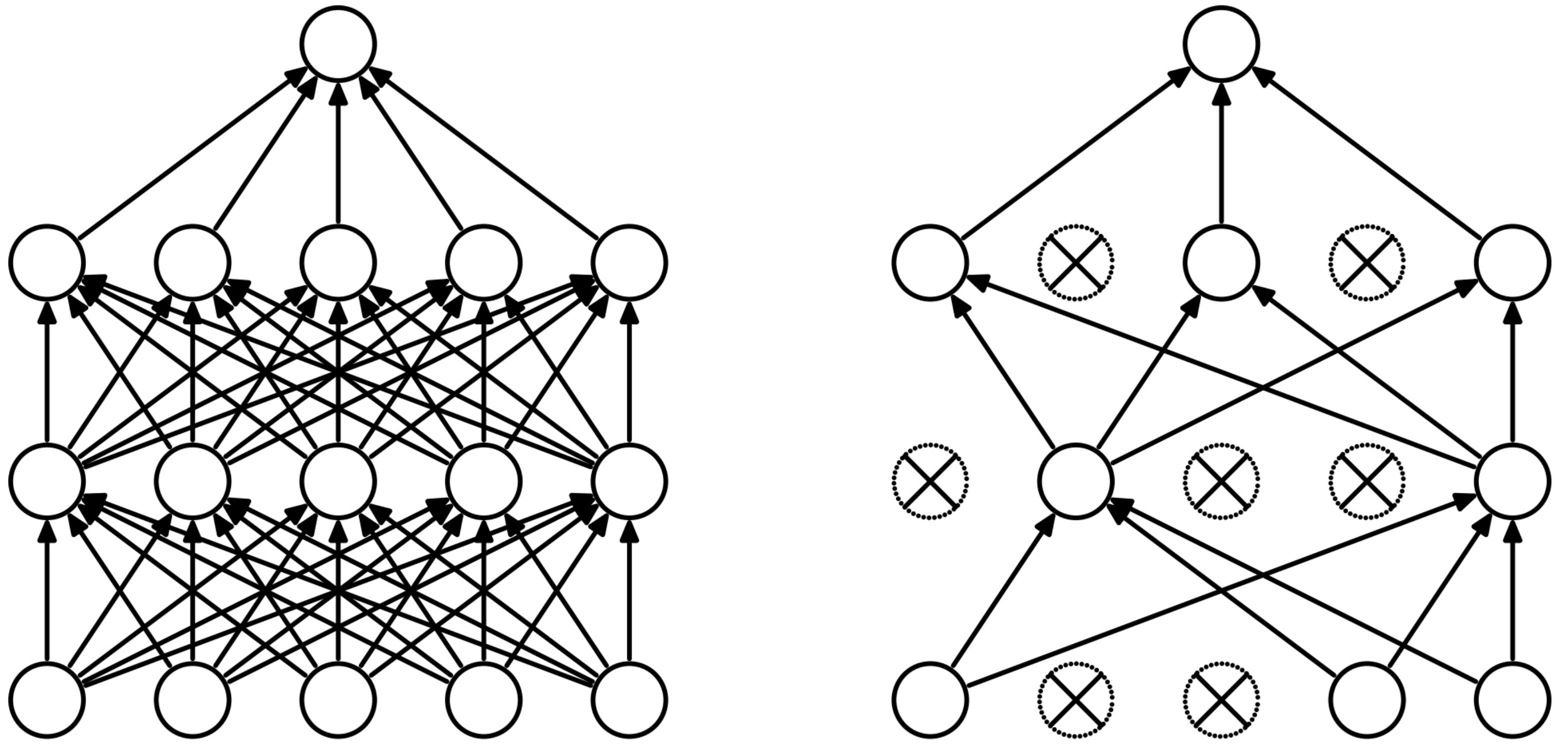
- Gaussian noise: Equivalent to minimizing loss with regularization term
- Encourages smooth function: small perturbation in weights leads to small changes in output

Injecting noise in output labels

- Better convergence: prevents pursuit of hard probabilities

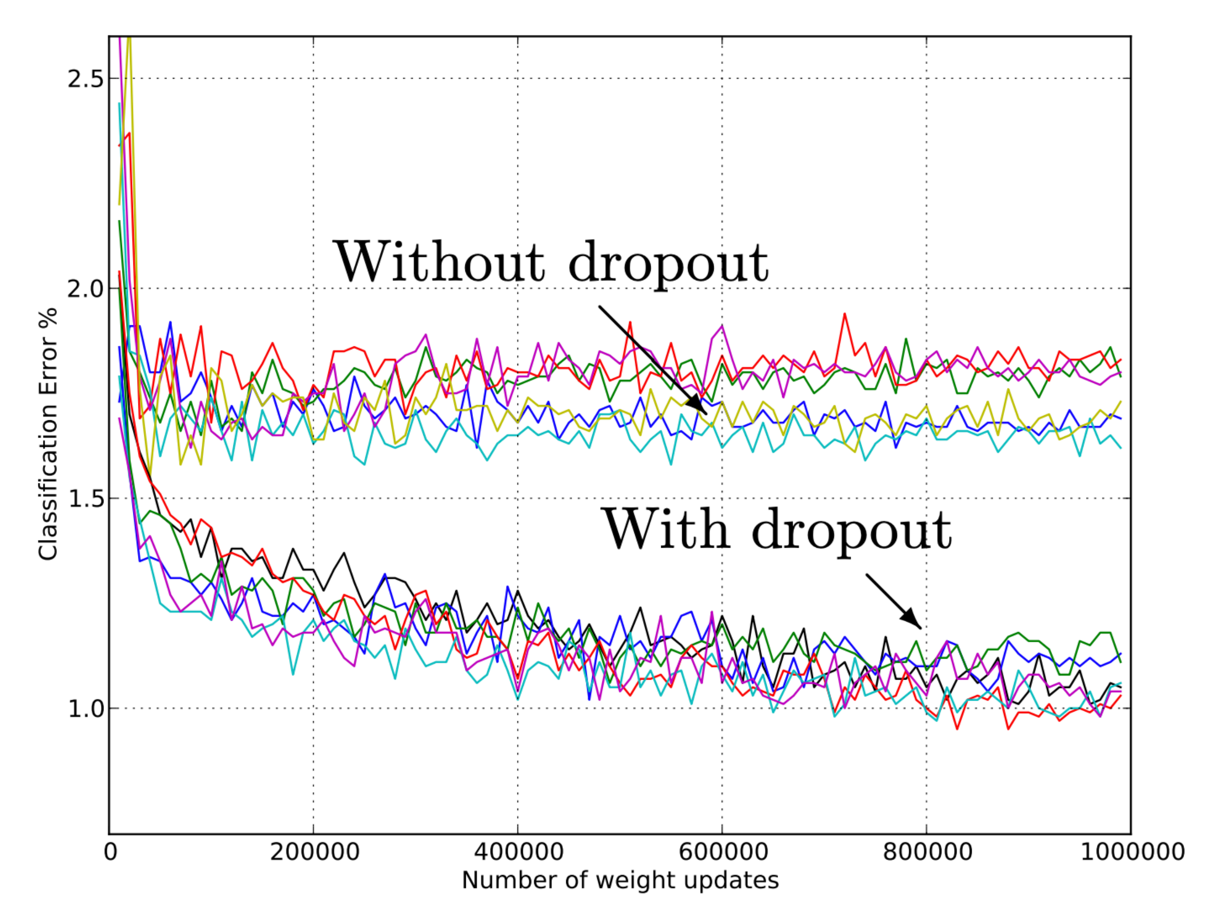
Dropout

- Randomly set some neurons and their connections to zero (i.e. "dropped")
- Prevent overfitting by reducing co-adaptation of neurons
- Like training many random sub-networks


(a) Standard Neural Net (b) After applying dropout.

Dropout

- Widely used and highly effective
- Proposed as an alternative to ensembling, which is too expensive for neural nets



Test error for different architectures with and without dropout. The networks have 2 to 4 hidden layers each with 1024 to 2048 units.

Image Source:
http://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf

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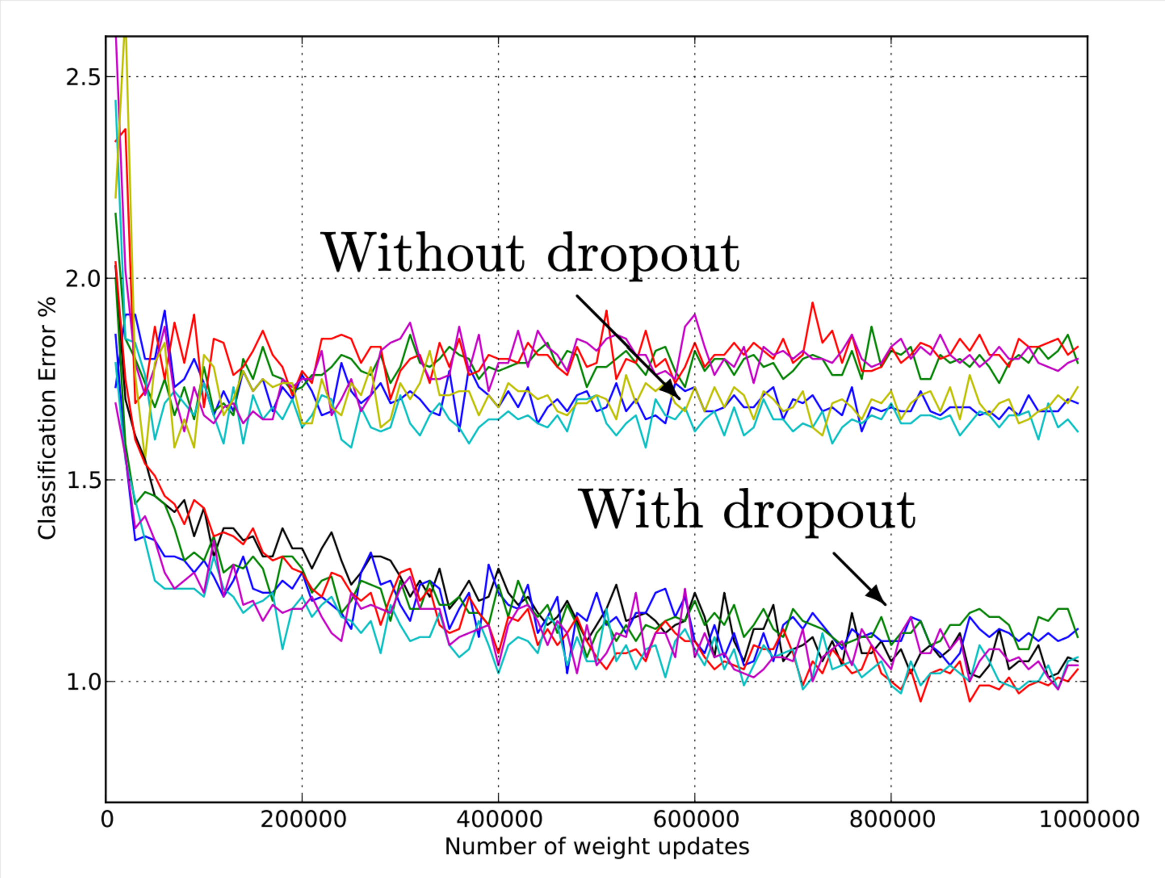

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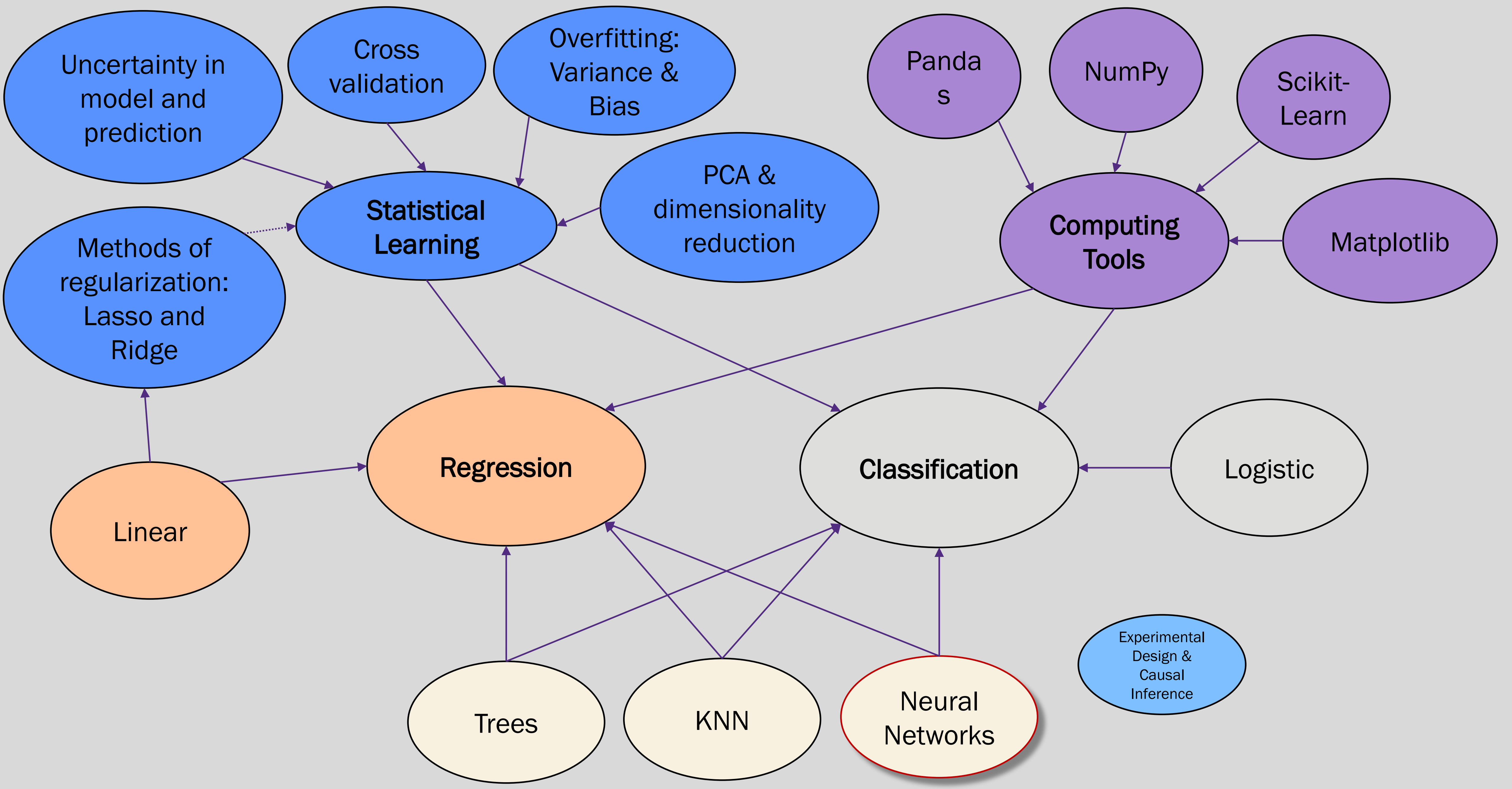
Uncertainty in model and prediction
Cross validation
Overfitting: Variance & Bias
Panda s
NumPy
Scikit-Learn
Statistical Learning
PCA & dimensionality reduction
Computing Tools
Matplotlib
Methods of regularization: Lasso and Ridge
Regression
Classification
Logistic
Linear
Trees
KNN
Neural Networks
Experimental Design & Causal Inference

Questions?